UAV-Based Remote Sensing of Soil Moisture Across Diverse Land Covers: Validation and Bayesian Uncertainty Characterization

Runze Zhang, Ishfaq Aziz, Derek A. Houtz, *Member, IEEE*, Yuxiang Zhao, Trent W. Ford, Adam C. Watts, and Mohamad Alipour, *Member, IEEE*

***Abstract*—High-resolution soil moisture (SM) observations are critical for agricultural monitoring, forestry management, and hazard prediction, yet current satellite passive microwave missions are unable to directly provide retrievals at tens-of-meter spatial scales. Unmanned aerial vehicle (UAV)–mounted microwave radiometry presents a promising alternative, but most evaluations to date have focused on agricultural settings, with limited exploration across other land covers and few efforts to quantify retrieval uncertainty. This study addresses both gaps by evaluating SM retrievals from a drone-based portable L-band radiometer (PoLRa) across shrubland, bare soil, and forest strips in Central Illinois, U.S., using a ten-day field campaign in 2024. Controlled UAV flights at altitudes of 10, 20, and 30 m were performed to generate brightness temperatures (T_B) at spatial resolutions of 7, 14, and 21 m. SM retrievals were carried out using multiple tau–omega-based algorithms, including the single-channel algorithm (SCA), the dual-channel algorithm (DCA), and the multitemporal DCA (MT-DCA). A Bayesian inference framework was then applied to provide probabilistic uncertainty characterization for both SM and vegetation optical depth (VOD). Results show that the gridded T_B distributions consistently capture dry–wet gradients associated with vegetation density variations, and spatial correlations between polarized observations are largely maintained across scales. Validation against in situ measurements indicates that PoLRa-derived SM retrievals from the SCA-V and MT-DCA algorithms achieve unbiased root mean square errors (ubRMSEs) generally below 0.04 m^3/m^3 across different land covers. Bayesian posterior analyses confirm that reference SM values largely fall within the derived uncertainty intervals, with mean uncertainty ranges around ± 0.02 and ± 0.11 m^3/m^3 for SCA- and DCA-related retrievals. These findings underscore the potential of UAV-mounted PoLRa for high-resolution SM retrieval across varied landscapes and emphasize the need for standardized calibration and uncertainty quantification frameworks to support broader scientific and operational adoption.**

***Index Terms*—Bayesian inference, multiresolution analysis, retrieval uncertainty, soil moisture (SM), unmanned aerial vehicle (UAV)-based radiometry.**

Received 30 May 2025; revised 20 October 2025; accepted 30 November 2025. Date of publication 4 December 2025; date of current version 17 December 2025. This work was supported by United States Department of Agricultural Forest Service under Grant 110789. *(Corresponding author: Runze Zhang.)*

Runze Zhang, Ishfaq Aziz, Yuxiang Zhao, and Mohamad Alipour are with the Department of Civil and Environmental Engineering, University of Illinois Urbana–Champaign, Urbana, IL 61801 USA (e-mail: runze@illinois.edu; ishfaqa2@illinois.edu; zhao132@illinois.edu; alipour@illinois.edu).

Derek A. Houtz is with the Microwave Remote Sensing, Swiss Federal Research Institute for Forest, Snow, and Landscape Research, 8903 Birmensdorf, Switzerland (e-mail: derek.houtz@wsl.ch).

Trent W. Ford is with Illinois State Water Survey, Prairie Research Institute, University of Illinois Urbana-Champaign, Champaign, IL 61820 USA (e-mail: twford@illinois.edu).

Adam C. Watts is with the Pacific Wildland Fire Sciences Laboratory, United States Forest Service, Wenatchee, WA 98801 USA (e-mail: adam.watts@usda.gov).

This article has supplementary downloadable material available at https://doi.org/10.1109/TGRS.2025.3640553, provided by the authors.

Digital Object Identifier 10.1109/TGRS.2025.3640553

I. Introduction

SOIL moisture (SM) plays a foundational role in hydrology and the land surface water cycle, serving as either an initial condition or a boundary condition in a wide range of hydrological models [1]. Accurate monitoring of SM dynamics is essential for advancing our fundamental understanding of land–atmosphere interactions and hydrological processes, thereby supporting progress across numerous Earth and environmental science disciplines [2], [3], [4]. In addition, previous studies have demonstrated that timely and precise SM information can significantly improve a variety of applications, including weather forecasting, drought monitoring, agricultural productivity assessments, food security prediction, and climate modeling [5], [6], [7], [8]. Given the importance of SM for both fundamental scientific research and practical applications, there is a critical need for observations that offer SM of both high precision and scalable spatial resolutions.

Microwave radiometry at lower frequencies, particularly at L-band (~1.4 GHz), offers significant advantages for SM monitoring due to the strong dielectric contrast between water and soil. Compared to C- and X-bands, L-band measurements exhibit superior atmospheric penetration, reduced vegetation attenuation, and greater sensing depth (~5 cm) [9]. The accuracy and applicability of satellite-based L-band SM products have been extensively validated across a variety of studies [10], [11], [12], [13], [14]. However, their coarse spatial resolutions, typically on the order of tens of kilometers, are primarily suited for large-scale applications such as numerical weather prediction and climate modeling [15]. Currently, while SM observations span scales from in situ point measurements to satellite footprints (~40 km), there remains a critical gap at intermediate scales (particularly from tens to hundreds of meters), significantly constraining the ability to capture localized variability and spatial hierarchies of SM.

To address these resolution gaps, aircraft-based L-band radiometers have been used to provide SM with both high accuracy and fine spatial resolutions [16], [17], [18]. However, airborne campaigns typically require complex logistics and

© 2025 The Authors. This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

entail substantial costs. As an alternative, various spatial downscaling schemes, such as the trapezoid model, have been extensively studied and applied to passive microwave SM products, aiming to preserve the accuracy of radiometer-based retrievals while capturing fine-scale spatial variability [19], [20], [21]. Nevertheless, these downscaled products often face significant validation challenges due to the scarcity of high-resolution reference data that can simultaneously provide an accurate magnitude of soil water content and correctly represent the spatial patterns of SM variations nested within the coarse-scale pixel units. In addition, the risk of introducing errors from ancillary datasets limits the reliability of downscaled SM fields, suggesting that directly capturing high-resolution SM from microwave observations remains the optimal strategy at present. Although spaceborne active microwave sensors, such as synthetic aperture radar (SAR), can deliver higher spatial resolution, their limited revisit frequency (~12 days), heightened sensitivity to surface roughness changes, higher operating frequency, and lower effective measurement depth, partially restrict their effectiveness for applications requiring timely and accurate SM retrievals [1], [22].

In recent years, the combination of unmanned aerial vehicles (UAVs) and microwave radiometers has emerged as a promising approach to capture SM at scalable resolutions, particularly ranging from a few meters to several hundred meters [23], [24], [25]. UAV-mounted L-band radiometers offer not only adjustable high spatial resolutions but also retrieval accuracy, cost-effectiveness, and flexible observation scheduling. The portable L-band radiometer (PoLRa), developed by TerraRad Tech, represents a novel, compact L-band radiometer specifically designed for deployment on various mobile platforms [26]. Unlike other small-scale L-band radiometers, such as the lobe differencing correlation radiometer (LDCR) and the airborne radiometer at L-band (ARIEL), which are limited to single-polarization measurements, PoLRa is capable of acquiring off-nadir brightness temperatures (T_B) at both vertical and horizontal polarizations [26], [27], [28]. SM retrievals based on PoLRa installed on pole-, rail-, and vehicle-fixed platforms have demonstrated their ability to capture SM variations over bare soils and grasslands at meter-scale resolutions with reasonable accuracy [29], [30], [31]. However, compared to ground-based deployments, UAV-based PoLRa SM retrievals introduce additional challenges, such as quickly changing incidence angles and footprint scaling with height above ground, which have not yet been systematically evaluated.

Hence, the primary objective of this study is to provide a comprehensive evaluation of UAV-based PoLRa-derived SM estimates. The urgent need to quantify topsoil water content for agricultural management has led most validation studies of drone-based passive microwave SM retrievals to focus on croplands [23], [25]. Many hydrometeorological applications, such as forest harvest and drought prediction, require SM information across diverse vegetation conditions. Therefore, in this study, the performance of PoLRa-derived SM retrievals was assessed across natural shrublands, bare soils, and mixed forests during a ten-day field campaign conducted in Central Illinois. Following experiment protocols proposed by Jackson [18], the spatial consistency of SM across different resolutions was examined at three separate altitudes: 10, 20, and 30 m. In addition, this study uniquely integrates surface temperature measurements acquired from the thermal infrared sensors onboard the Da-Jiang Innovations (DJIs) Mavic 3T [32]. By separately considering canopy temperatures from thermal infrared observations and soil temperatures from in situ probes at approximately 5-cm depth, the retrieval framework becomes independent of the thermal equilibrium assumption typically valid for early morning observations [9]. This approach lays the groundwork for extending drone-based SM sensing beyond the conventional 6 A.M. observation window, enabling flexible diurnal measurements and facilitating the coordinated use of multiple UAV-mounted sensors for SM sensing.

Given the increasing need for real-time vegetation monitoring and the potential to infer fuel moisture from microwave-derived vegetation optical depth (VOD), the utilization of dual-polarized observations for simultaneously retrieving SM and VOD has become an area of growing interest [33], [34]. However, the conventional dual-channel algorithm (DCA) is susceptible to large retrieval biases because of minor observational noises, imperfect forward models, compensatory effects during optimization, and the strong dependence between polarized observations [35], [36]. To mitigate these issues, the multitemporal DCA (MT-DCA) was developed, enabling an overdetermined retrieval system where the number of observations exceeds the number of unknown parameters [37], [38]. A simplified version of the MT-DCA approach was incorporated in this study to preliminarily assess the utility potential of UAV-mounted microwave radiometer data.

Beyond providing a comprehensive evaluation of PoLRa-derived SM retrieval across different land covers, resolutions, algorithms, and thermal assumptions, a secondary objective of this work is to estimate the retrieval uncertainty through the coupled application of Bayesian inference and Markov chain Monte Carlo (MCMC) simulations [39], [40]. The application of probabilistic distributions as uncertainty information for SM and VOD retrievals has not been widely explored, especially when compared to extensive efforts focused on their deterministic error characterizations. Previous uncertainty characterization efforts often involve assigning fixed ranges to input parameters and observations; these methods treat all values within the range as equally probable, making the results highly sensitive to initial guesses [17], [35]. Furthermore, nonprobabilistic inversions may not represent complex multimodal solutions or inversion ambiguities resulting from noisy real-world data. In contrast, Bayesian inference yields full posterior distributions that integrate both model-data mismatch and inversion ambiguities, providing a more robust representation of retrieval uncertainty. This study primarily focuses on demonstrating the feasibility of applying Bayesian inference to SM retrievals in the context of a zeroth-order radiative transfer model. The inclusion of credible intervals around the retrieval estimates further enhances the interpretability and reliability of the resulting SM and VOD products.

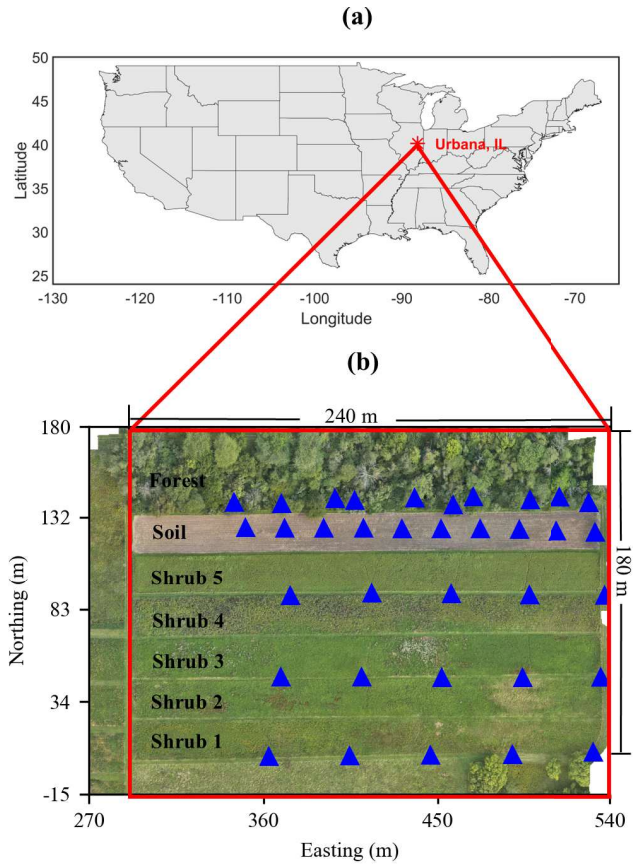


Fig. 1. (a) Geographic location of Urbana, IL, on the U.S. map. (b) Layout of the experimental site highlighted by the red box for evaluating SM retrievals from the drone-mounted PoLRa, consisting of seven parallel strips covering forest, bare soil, and natural shrubland. Strip names are marked using the bold font. Blue triangles denote the locations of in situ probe measurements.

II. Experiment Description

All field experiments were conducted at the Phillips Tract Research Plots within the University of Illinois Natural Areas, Urbana, IL [see Fig. 1(a)]. The study site spans approximately 240 × 180 m and features seven parallel strips with varying land cover types, including shrublands, bare soil, and mixed forest [see Fig. 1(b)]. Shrub strips are managed under different vegetation clearing frequencies, ranging from annual to multiple-year intervals, resulting in distinct land cover characteristics despite comparable vegetation densities. For clarity, shrub strips are identified by a combination of the term "Shrub" and their respective strip order (i.e., Shrub 1 for the bottommost strip and Shrub 5 adjacent to bare soil) [see Fig. 1(b)]. In situ measurements of SM and soil temperature at a 5-cm depth were collected at 35 fixed locations, marked with blue triangles on the map, using METER TEROS12 probes [41]. These 35 points were evenly distributed across forest, soil, and transitional shrubland areas. To ensure measurement consistency, each location was permanently marked with red flags, and three replicate measurements were taken around each flag and averaged to represent local conditions for validation. However, due to accessibility constraints, several measurements were collected along pathways between shrub strips, potentially limiting their representativeness of adjacent strips. The experimental campaign was conducted from September 4 to 13, 2024, with flights and measurements restricted to a 9:30–11:30 A.M. window each day to minimize diurnal variability induced by temperature and evapotranspiration. Experiments on September 13 were curtailed due to unexpected rainfall. Despite this, soil samples had been collected on the final day (prior to precipitation) for gravimetric water content measurements and soil texture analysis, revealing an average clay fraction of 8.5% across the site.

To minimize battery replacements and optimize flight efficiency, PoLRa flights were conducted in two batches: 20 and 30 m (first batch) and 10 m (second batch). The 30-m flight altitude was selected based on maximum tree heights (of around 26 m) in the forest strip. Therefore, observations at 10 and 20 m do not include the contribution from forested areas. PoLRa scanned each strip from right to left and back (i.e., left to right) before moving to the next, with flight paths designed to maximize coverage of homogeneous areas while avoiding vegetation contact. Given PoLRa 3-dB full beamwidth of 37°, footprint dimensions along and across track were calculated, accounting for beam position correction and incidence angle variations. Although PoLRa is designed for observation at a nominal 40°, actual observation angles could span from 20° to 60°. Only data with incidence angles between 30° and 50° were retained for further analysis. To standardize spatial resolutions, conservative circular footprints were approximated at 7-, 14-, and 21-m diameters for flight altitudes of 10, 20, and 30 m, respectively [see Fig. 2(b)]. The chosen spatial resolutions were determined by the lengths of the minor axes of the elliptical footprints [i.e., the "height" in Fig. 2(b)]. Given the width of each strip, the use of these along-track sizes [i.e., the "width" in Fig. 2(b)] as spatial resolutions might lead to a misunderstanding that an individual emission observed over a strip is partly contributed by its adjacent strip, particularly with a flight elevation of 30 m. We acknowledge that approximating the true elliptical footprints as circular footprints neglects partial contributions from the outer elliptical zones and may, therefore, introduce some bias. However, because along-strip conditions exhibit relatively homogeneous, gently varying gradients and because the scene was heavily oversampled at 14 Hz with an overall flight speed around 7 m/s, the incremental contribution of the elongated elliptical "tails" extending beyond the circular boundary is expected to be minor.

In addition, a DJI Mavic 3T equipped with visible and infrared sensors was flown daily during the campaign, providing surface reflectance and canopy temperature data. The canopy temperatures estimated from Mavic 3T observations over vegetated strips were incorporated into the SM retrieval process to reduce dependence on the thermal equilibrium assumption, which is typically valid only near 6 A.M.

III. Methodology

A. Sky Measurement Radiometer Characterization

At microwave frequencies, the intensity of the observed emission is commonly represented by brightness

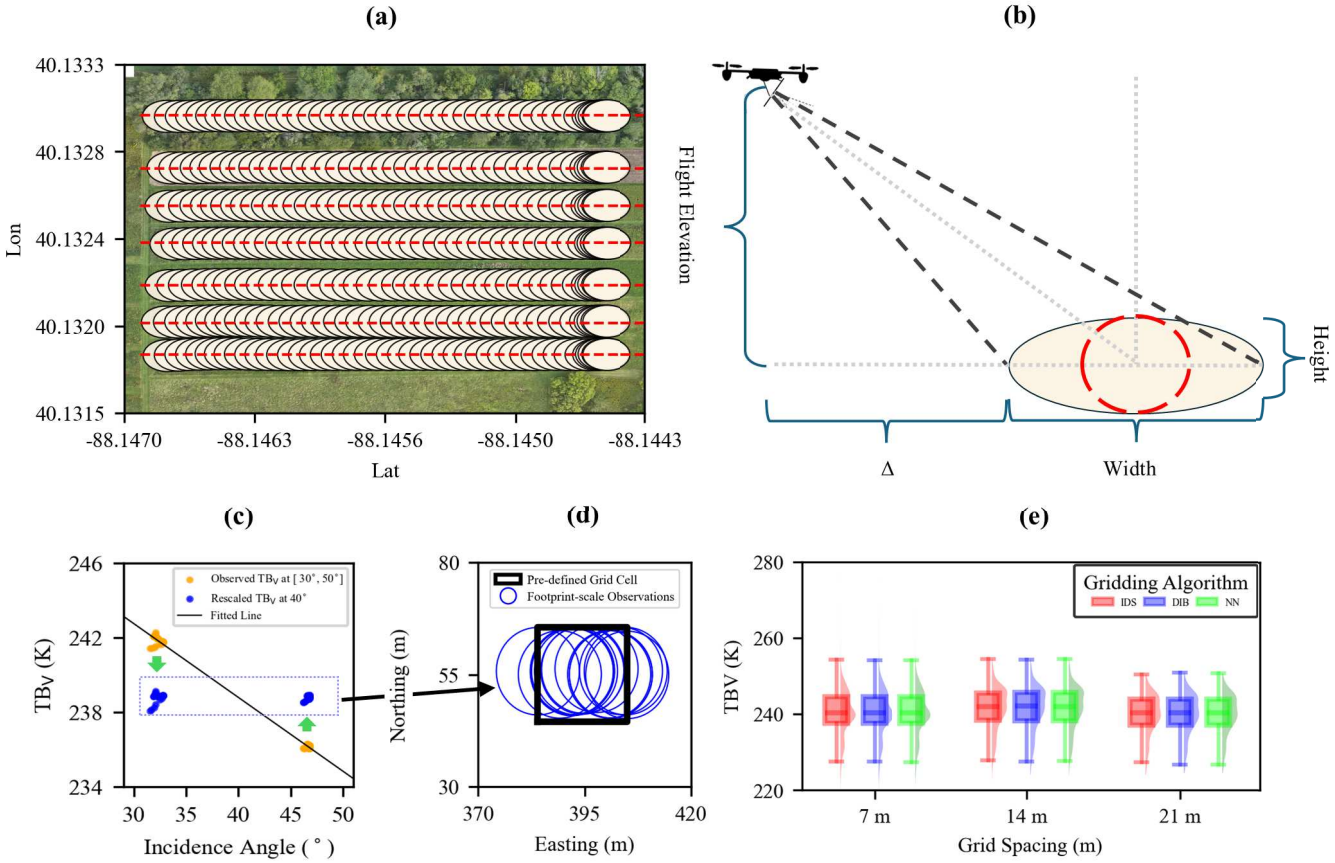


Fig. 2. (a) Spatial distribution of PoLRa 3-dB footprints along flight tracks at an altitude of 30 m. (b) Conceptual geometry of one PoLRa 3-dB footprint-scale observation at an incidence angle of around 40°, where the red circle delineates the strategy for determining spatial resolution. (c) Example of angular normalization (to 40°) of the footprint-scale vertically polarized brightness temperatures with incidence angles from 30° to 50°. This linear regression is conducted using all the footprint-scale observations falling within the same predefined grid cell. The rescaled brightness temperatures (blue dots) are used to generate the representative brightness temperature of the grid cell. (d) Schematic illustrating how multiple footprint-scale vertically polarized brightness temperatures (blue circles) are present within a single predefined grid cell (black square). (e) Boxplots of vertically polarized brightness temperatures at gridded scales of 7, 14, and 21 m using three different spatial interpolation methods: IDS, DIB, and nearest neighbor (NN).

temperature (T_B). In the context of PoLRa, quantifying the surface-emitted energy captured by the antenna ($T_{B_p}^{\text{ant}}$) begins with converting the raw voltage readings at the power detector (u_p) into the brightness temperature at the switch input ($T_{B_p}^{\text{in}}$), which requires knowledge of radiometer gain (G) and inherent offset noise temperature ($T_{B_{\text{off}}}$) (1). Here, the subscript p refers to either horizontal (H) or vertical polarization (V). Both G and $T_{B_{\text{off}}}$ are not fixed over time and are dynamically determined based on the relative changes between two internal reference measurements: the active cold source (ACS) and the ambient matched resistive source (RS) within the PoLRa. Using these internal references, G and $T_{B_{\text{off}}}$ can be calculated synchronously with $T_{B_p}^{\text{in}}$, as in (2) and (3). Given that the magnitudes of $T_{B_{\text{RS}}}$ and $T_{B_{\text{ACS}}}$ depend on the physical temperature of the internal reference sources, and the PoLRa hardware is not temperature-stabilized, it is essential to characterize the temperature dependence of $T_{B_{\text{RS}}}$ and $T_{B_{\text{ACS}}}$. For RS reference, $T_{B_{\text{RS}}}$ is usually assumed equivalent to its physical temperature (T_{RS}), measured directly via an internal sensor. In contrast, $T_{B_{\text{ACS}}}$ is expected to vary linearly with its physical temperature (T_{ACS}) (4) [42], [43]

$$T_{B_p}^{\text{in}} = G \cdot u_p + T_{B_{\text{off}}} \tag{1}$$

$$G = \frac{T_{B_{\text{RS}}} - T_{B_{\text{ACS}}}}{u_{\text{RS}} - u_{\text{ACS}}} \tag{2}$$

$$T_{B_{\text{off}}} = -G \cdot u_{\text{RS}} + T_{B_{\text{RS}}} \tag{3}$$

$$T_{B_{\text{ACS}}} = m \cdot T_{\text{ACS}} + b. \tag{4}$$

It is important to note that the yielded $T_{B_p}^{\text{in}}$ comprises both: 1) the portion of external radiation entering the antenna that reaches the switch input ($(1-\alpha_p)\cdot T_{B_p}^{\text{ant}}$) and 2) internal emissions from the transmission path (TP) composed of the antenna and cable ($\alpha_p \cdot T_{\text{TP}}$) (5). $T_{B_p}^{\text{ant}}$ represents the brightness temperature incident on the antenna at polarization p, primarily contributed by the target surface, which will serve as the observation inputs for SM and VOD retrievals. The parameter α_p is the absorption coefficient of the antenna-cable system and is derived from L_p, which denotes the cumulated loss between the antenna and switch input in decibels (dB) (6), resulting from nonideal antenna efficiency, cable losses, and

TABLE I
POLRA CALIBRATION PARAMETERS (m, b, AND L) USED FOR THE ACS CHARACTERIZATION AND DAILY POLARIZED TEMPERATURE OFFSETS (d_p)

Applicable Period	Parameters		
	m	b (K)	L (dB)
September 4 – 12, 2024	0.355	-90.000	-0.100
Applicable Date	d_H (K)	d_V (K)	
September 4, 2024	-52.478	-55.992	
September 5, 2024	-63.117	-59.058	
September 6, 2024	-66.694	-57.914	
September 7, 2024	-63.236	-51.538	
September 8, 2024	-65.903	-48.817	
September 9, 2024	-83.946	-63.600	
September 10, 2024	-62.923	-51.155	
September 11, 2024	-62.688	-50.287	
September 12, 2024	-68.990	-59.229	

connector losses [26], [29]. In this study, polarization effects on antenna-cable losses are assumed negligible ($L_H = L_V = L$), and T_{TP} is approximated as the average of the antenna (T_{ant}) and cable temperatures (T_{cab}) (7).

With voltage and temperature measurements, $T_{B_p}^{\mathrm{ant}}$ can be derived once the parameters m, b, and L are known (8). Calibration of these ACS-related parameters (m, b, and L) is typically achieved by introducing an external emission with known T_B values and observing this known source over a range of physical temperatures. Sky and water surfaces are often used as reliable references for calibrating low-frequency radiometers [18]. At L-band, the sky T_B, including extraterrestrial radiation, is often on the order of several Kelvin [44], [45]. Assuming a sky T_B of 5 K and using voltage measurements from PoLRa pointed at the clear sky, m, b, and L values can be determined via an iterative optimization procedure that minimizes the differences between $T_{B_p}^{\mathrm{ant}}$ (sky) and the expected 5 K value. In this study, it should be noted that m, b, and L parameters were directly adopted from prior calibration performed on the same PoLRa instrument (see Table I) [30]. Daily sky-pointing measurements during the field campaigns were additionally used to compute daily polarized temperature offsets (d_p), partially compensating for systematic deviations in $T_{B_p}^{\mathrm{ant}}$ likely caused by the unexpected radiometer frequency interference (RFI) contamination (see Table I). Notably, the correction illustrated by (8) is only designed for this work, with the specific reasons for its adoption provided in Section IV-D

$$T_{B_{\mathrm{P}}}^{\mathrm{in}} = \left(1-\alpha_p\right)\cdot \mathrm{T}_{B_p}^{\mathrm{ant}} + \alpha_p \cdot T_{\mathrm{TP}} \tag{5}$$

$$\alpha_p = 1-10^{\frac{L_p}{10}} \tag{6}$$

$$T_{\mathrm{TP}} = \frac{T_{\mathrm{ant}} + T_{\mathrm{cab}}}{2} \tag{7}$$

$$T_{B_{\mathrm{P}}}^{\mathrm{ant}} = \frac{T_{B_{\mathrm{P}}}^{\mathrm{in}} - \alpha_p \cdot T_{\mathrm{TP}}}{\left(1-\alpha_p\right)} + d_p. \tag{8}$$

B. Spatial Gridding of Footprint-Scale Brightness Temperatures

Along the prescribed flight paths, the PoLRa's 3-dB elliptical footprints at a flight altitude of 30 m have been calculated and illustrated in Fig. 2(a). With an integrated sampling cycle of 69 ms, the time required to sequentially record one round of voltage and temperature measurements at the ACS, RS, H-pol, and V-pol switches, PoLRa generates approximately 14 measurement sets per second, corresponding to 28 $T_{B_p}^{\mathrm{ant}}$ values. Given this high observation density, footprint-level retrievals not only demand substantial computational resources but are also more susceptible to random noises. Furthermore, substantial overlap among adjacent footprints results in oversampling (of redundant information), limiting the contribution of unique observational content. Therefore, footprint-scale observations are commonly aggregated onto spatial grids, producing effective brightness temperature per pixel ($T_{B_p}^{\mathrm{grid}}$) [46]. Following this approach, $T_{B_p}^{\mathrm{ant}}$ samples collected at flight altitudes of 10, 20, and 30 m were grouped, filtered, and gridded at corresponding pixel resolutions of 7, 14, and 21 m. Before gridding, the latitude and longitude of each beam center were corrected based on attitude angles and reprojected into the Universal Transverse Mercator (UTM) coordinate system. Observations with incidence angles outside 30°–50° were excluded from further analysis. Removing measurements in the [20°, 30°] and [50°, 60°] ranges keeps the dataset within ±10° of the instrument nominal operating angle, limiting footprint inflation, and trimming the small tails most affected by wind-induced roll. Notably, in this study, the aggregated $T_{B_p}^{\mathrm{ant}}$ values computed from individual footprints restricted to [30°, 50°] are closely comparable to those computed from the broader [20°, 60°] (see Fig. S1).

To mitigate bias in the derived $T_{B_p}^{\mathrm{grid}}$ from varying $T_{B_p}^{\mathrm{ant}}$ values because of wind-affected viewing geometries, all $T_{B_p}^{\mathrm{ant}}$ observations within a given 7-, 14-, or 21-m grid cell were first normalized to a standard 40° incidence angle. This normalization was achieved by applying a linear regression to model the relationship between $T_{B_p}^{\mathrm{ant}}$ observations and their incidence angles [see Fig. 2(c)]. The clustering of $T_{B_p}^{\mathrm{ant}}$ values into two groups surrounding 40° occurred because the sensor consecutively scanned the same strip twice from opposite directions. After normalization, the rescaled $T_{B_p}^{\mathrm{ant}}$ values were aggregated into a single effective $T_{B_p}^{\mathrm{grid}}$ for the grid [see Fig. 2(d)]. Three methods were applied to derive the $T_{B_p}^{\mathrm{grid}}$ value, which are 1) drop-in-bucket (DIB) method; 2) NN method; and 3) inverse-distance-squared (IDS) method [47]. For DIB, the arithmetic mean of $T_{B_p}^{\mathrm{ant}}$ samples within a grid cell is assigned as the representative $T_{B_p}^{\mathrm{grid}}$, whereas NN directly selects the footprint $T_{B_p}^{\mathrm{ant}}$ closest to the center of the grid cell as $T_{B_p}^{\mathrm{grid}}$. Regarding the IDS interpolation, the effective $T_{B_p}^{\mathrm{grid}}$ value is calculated as a weighted average of all the footprint-scale $T_{B_p}^{\mathrm{ant}}$ data within the grid cell, where the weights are inversely proportional to the square of the distance between each boresight location and the grid cell center (9). Fig. 2(e) shows that the resulting distributions of $T_{B_p}^{\mathrm{grid}}$ values are similar across these methods. Given that the IDS interpolation offers

a favorable balance between resolution degradation and noise reduction, it was selected for subsequent SM analyses [46]. However, it is important to note that interpolation does not fully eliminate biases in $T_{B_p}^{\text{grid}}$ values. In addition to random instrument noise, residual radiation contribution from regions outside the nominal 3-dB footprints remains a potential source of error

$$T_{B_p}^{\text{grid}} = \frac{\sum_{i=1}^{N} \beta_i T_{B_p}^{\text{ant}_i}}{\sum_{i=1}^{N} \beta_i}$$
$$\beta_i = \frac{1}{d_i^2}$$
$$d_i = \sqrt{\left(x_i - x^{\text{grid}}\right)^2 + \left(y_i - y^{\text{grid}}\right)^2} \tag{9}$$

where the subscript i denotes the ordinal number of one snapshot within the predefined grid cell. x_i and y_i represent the coordinates of the beam center of the ith snapshot.

C. τ–ω Semiempirical Model

The τ–ω model, a zeroth-order radiative transfer model, simulates microwave thermal emissions from the land surface in the form of T_B by accounting for vegetation attenuation effects [48]. Neglecting the minor contribution of reflected downwelling atmospheric and extraterrestrial radiation, the upwelling emission from the land surface can be represented by three components, which are: 1) upward soil emission attenuated by the vegetation layer; 2) direct vegetation emission; and 3) vegetation emission reflected off the soil surface (11) [9]. Conventional SM retrieval is performed by minimizing the differences between observed and simulated T_B values through iteratively updating the input variables, i.e., SM and VOD (or τ). The forward simulation begins by converting an estimated volumetric water content into dielectric permittivity using a dielectric mixing model. The Mironov model is applied in this study [49]. Polarized surface reflectivity (r_{s_p}) is then calculated from the dielectric constant via the Fresnel equations. To account for surface roughness effects, the H-Q-N semiempirical roughness model is used to derive the rough surface reflectivity (r_{r_p}) (12) [50]. The radiative contribution of the canopy layer is incorporated, with attenuation strength determined by τ, a composite parameter influenced by vegetation structures, water content, biomass, and canopy temperature [51], [52].

The specific objective functions for retrieval depend on the chosen algorithm and the targeted variables. In the classical single-channel algorithm (SCA), SM is retrieved from a single polarization T_B, while τ is estimated independently using multispectral visible light vegetation indices, such as the normalized difference vegetation index (NDVI), as in (16) and (17) [9]. In DCA, both H-pol and V-pol T_B measurements are used simultaneously, allowing the retrieval of both SM and τ by minimizing the sum of squared residuals between observation and simulated T_{B_p} values (14). However, DCA retrievals are often highly sensitive to observational noises and model imperfections [53]. This sensitivity is primarily due to the mutual dependence between dual-polarized T_B values within the same snapshot, which limits the degrees of independent information available for inversion [36]. To mitigate this issue, the MT-DCA was developed, using four T_B observations from two successive scans to retrieve three unknowns where SM keeps varying and τ is assumed to remain constant for scans from the two consecutive days (15) [37], [38]. Given that MT-DCA has rarely been evaluated for UAV-mounted L-band radiometers that capture high-resolution variations and may suffer from calibration challenges, this study adopts MT-DCA alongside traditional SCA and DCA approaches based on the same set of observations. However, relative to the original MT-DCA, the implementation in this study has been simplified by prescribing a fixed value for the single scattering albedo (ω). Specifically, ω is set as 0.08 for vegetated surfaces, corresponding to the median value of the global ω map derived from SMAP MT-DCA outputs [37]. In addition, following [37], the roughness parameter (h) of 0.13 was applied across all land cover types. For consistency, these h and ω values have been uniformly utilized for all retrieval algorithms

$$\gamma = e^{-\frac{\tau}{\cos\theta}} \tag{10}$$
$$T_{B_p}^{\text{grid}} = \gamma\left(1 - r_{r_p}\right) T_s + (1-\gamma)(1-\omega) T_c + \gamma r_{r_p} (1-\gamma)(1-\omega) T_c \tag{11}$$
$$r_{r_p} = \left[i + (1-Q)\, r_{s_q}\right] e^{-h\cos^2\theta} \tag{12}$$
$$J_{\text{SCA}}\left(\text{sm}_{\text{ret}}\right) = \left[T_{B_p}^{\text{sim}}\left(\text{sm}_{\text{ret}}\right) - T_{B_p}^{\text{grid}}\right]^2 \tag{13}$$
$$J_{\text{DCA}}\left(\text{sm}_{\text{ret}}, \tau_{\text{ret}}\right) = \left[T_{B_v}^{\text{sim}}\left(\text{sm}_{\text{ret}}\right) - T_{B_v}^{\text{grid}}\right]^2 + \left[T_{B_H}^{\text{sim}}\left(\text{sm}_{\text{ret}}\right) - T_{B_H}^{\text{grid}}\right]^2 \tag{14}$$
$$J_{\text{MT-DCA}}\left(\text{sm}_{\text{ret}}^{t_1}, \text{sm}_{\text{ret}}^{t_2}, \tau_{\text{ret}}\right) = \sum_{i=1}^{2}\left[T_{B_v}^{\text{sim}}\left(\text{sm}_{\text{ret}}^{t_i}\right) - T_{B_v}^{\text{grid},t_i}\right]^2 + \left[T_{B_H}^{\text{sim}}\left(\text{sm}_{\text{ret}}^{t_i}\right) - T_{B_H}^{\text{grid},t_i}\right]^2. \tag{15}$$

D. Ancillary Datasets

In the τ–ω model, the roles of the canopy temperature (T_c) and the soil effective temperature (T_s) in retrieving SM from $T_{B_p}^{\text{grid}}$ measurements are explicitly shown in (11). Conventionally, these two temperatures are assumed to be equal for simplification (i.e., $T_s = T_c$). However, given that the experiments in this study were conducted between 9:30 A.M. and 11:30 A.M., a period during which canopy and soil temperatures typically diverge (i.e., $T_s \neq T_c$), the use of a single soil effective temperature to represent both near-surface soil and canopy temperatures is likely inappropriate [9], [54]. Therefore, a thermal infrared sensor onboard the DJI Mavic 3T, operating at wavelengths of 8–14 μm, was deployed to independently measure T_c, immediately prior to PoLRa observations, assuming minimal temperature fluctuation during the experimental window. At a flight altitude of 30 m, the unit pixel size of the Mavic 3T thermal images is approximately 0.1 m [see Fig. 3(d)]. These thermal maps were reprojected into predefined grid cells of 7, 14, and 21 m via the DIB approach [see Fig. 3(a)–(c)]. In addition, daily averages of

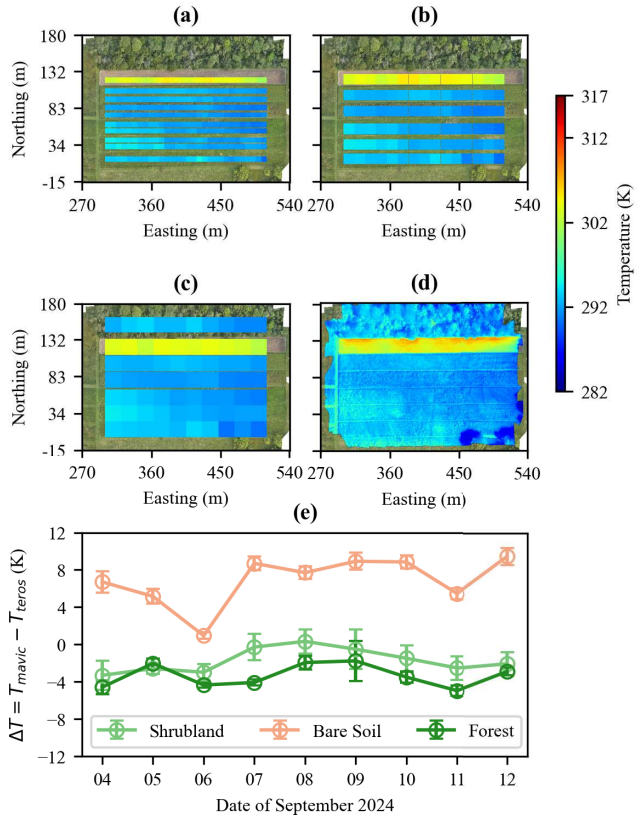


Fig. 3. Spatial distributions of canopy and soil surface temperatures measured by the DJI Mavic 3T at the grid cells of (a) 7, (b) 14, (c) 21, and (d) 0.1 m on September 8, 2024. (e) Time series of the mean temperature differences between the temperatures provided by Mavic 3T (T_{mavic}) and TEROS 12 (T_{teros}) across land covers. T_{mavic} is assumed to represent canopy temperature (T_c) in vegetated strips and soil skin temperature in the soil strip. Each point in the time series is an average of all temperature differences between the T_{teros} measurements and their corresponding 21-m gridded T_{mavic} values [see Fig. 3(c)] under the same land cover per day, and the standard deviations of daily temperature differences are displayed as the error bars. For shrubland, temperature differences are averaged over multiple strips as TEROS measurements are distributed at Shurb 1, Shrub 3, and Shrub 5, as shown in Fig. 1. For soil and forest, temperature differences are computed based on TEROS measurements within a single strip.

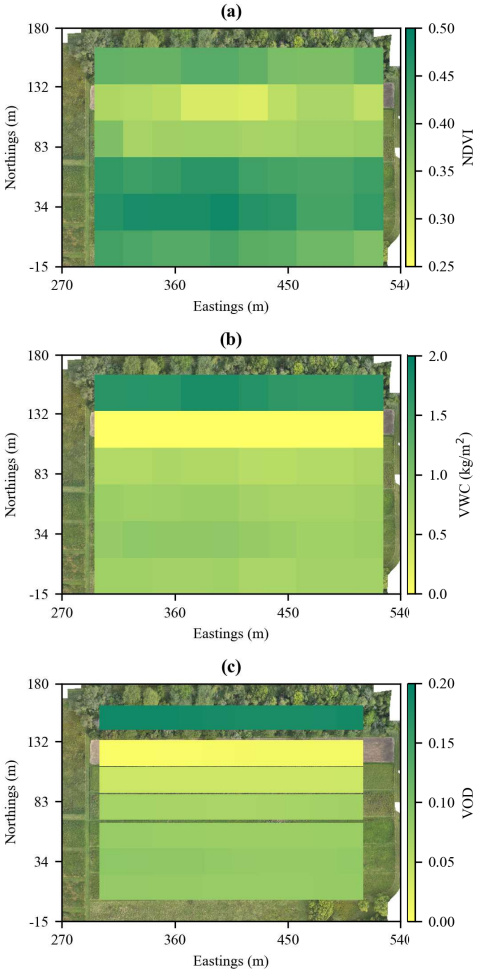


Fig. 4. Spatial maps of (a) Landsat 8 30-m NDVI, (b) NDVI-derived VWC, and (c) gridded VOD at a resolution of 21 m during the experimental period.

soil temperatures measured using the TEROS 12 probes' pins were used as T_s [55]. During the experimental period, the standard deviations of T_s were generally within 2 K. While TEROS12 temperature readings may not be representative of the full 5-cm soil layer, T_s values are systematically different from surface soil temperatures measured by Mavic 3T, reaching approximately 8 K [see Fig. 3(e)]. These large deviations could be partially caused by the flight height of 30 m that exceeds the manufacturer's recommended maximum distance of 25 m between the thermal camera and the ground surface (private communication with DJI support). Meanwhile, they also indicate that TEROS 12 sensors are still able to capture reliable soil temperatures below the surface. On the other hand, differences between T_c and T_s are relatively stable over time, typically ranging from 0 K to about −4 K [see Fig. 3(e)].

As mentioned above, accurate estimation of VOD (or τ) is imperative for SCA and is often derived from multispectral vegetation indices, including NDVI and the leaf area index (LAI) [51], [56]. In the SMAP retrieval framework, for example, VOD (or τ) is computed as the product of a vegetation-related parameter (b_{LC}) and vegetation water content (VWC), with VWC derived from a long-term NDVI climatology with land-cover-dependent coefficients (17) [9].

Although the use of NDVI climatology-based τ cannot fully capture real-time vegetation dynamics and can suppress short-term SM anomalies, the bulk precision of SMAP SCA SM retrievals remains comparable (or even better) to that of other algorithms [57], [58]. Given the short duration of this study's experimental period, the impact of using fixed NDVI-derived τ values on PoLRa SCA retrieval accuracy is expected to be limited. Surface reflectance data from red and near-infrared bands of Landsat 8 Level 2 and the Sentinel-2 Level 2 datasets (October 1, 2023–September 30, 2024) have been used to compute maximum, minimum, and effective NDVI values at 30- and 10-m resolutions [59], [60]. These NDVI maps were first converted into VWC and VOD using (16), and the derived VOD matrices were subsequently resampled through spatial averaging to match the target retrieval resolutions (see Figs. 4 and S2). Our internal analysis indicated that the PoLRa SCA SM retrievals based on Landsat 8 and Sentinel-2

TABLE II
EVALUATION METRICS OF POLRA-DERIVED SM RETRIEVALS USING DIFFERENT ALGORITHMS

Algorithm	Resolution (m)	Bias (m^3/m^3)	RMSE (m^3/m^3)	ubRMSE (m^3/m^3)	R
SCAV	7	0.088	0.094	0.031	0.257
	14	0.077	0.083	0.029	0.324
	21	0.103	0.110	0.036	0.207
SCAH	7	0.096	0.107	0.045	-0.146
	14	0.077	0.091	0.047	-0.144
	21	0.107	0.123	0.059	-0.181
DCA	7	0.100	0.110	0.044	0.451
	14	0.094	0.106	0.045	0.464
	21	0.114	0.127	0.054	0.404
MT-DCA	7	0.092	0.099	0.035	0.403
	14	0.089	0.098	0.037	0.441
	21	0.108	0.116	0.041	0.422

VOD products have similar accuracy to each other (see Table II and Table S1). To ensure clarity and avoid redundancy, SCA SM retrievals based on the Landsat 8 VOD product have been solely adopted for subsequent analyses

$$\mathrm{VWC} = \left(1.9134 \cdot \mathrm{NDVI}_{\mathrm{eff}}^2 - 0.3215 \cdot \mathrm{NDVI}_{\mathrm{eff}}\right) + f_{\mathrm{stem}} \cdot \frac{\mathrm{NDVI}_{\max} - \mathrm{NDVI}_{\min}}{1 - \mathrm{NDVI}_{\min}} \tag{16}$$

$$\tau = b_{\mathrm{LC}} \cdot \mathrm{VWC}. \tag{17}$$

E. Bayesian Inference and MCMC Simulation

Bayesian inference is a statistical approach that uses Bayes's theorem to update prior beliefs based on observed data, yielding posterior probabilities for unknown parameters [40], [61]. In this study, we used Bayesian inference to probabilistically estimate SM and VOD from observed brightness temperatures, i.e., $T_{B_p}^{\mathrm{grid}}$. According to Bayes' theorem, the posterior probability distribution, $\Pr(\theta|T_{B_p}^{\mathrm{grid}})$, of the unknown parameters, $\theta = [\mathrm{sm}, \mathrm{VOD}]$, is computed by updating prior knowledge $\Pr(\theta)$ with the likelihood function $\Pr(T_{B_p}^{\mathrm{grid}}|\theta)$ (18). In (18), the normalizing term $\Pr(T_{B_p}^{\mathrm{grid}})$ is commonly ignored, and thus, the posterior distributions of the unknown parameters can be inferred using (19). In this study, uniform priors within fixed pragmatically defined ranges of SM and VOD are used. The likelihood function is modeled in logarithmic form, assuming that the error between the radiometric simulations (i.e., $T_{B_p}^{\mathrm{sim}}$) and observations (i.e., $T_{B_p}^{\mathrm{grid}}$) exhibits a Gaussian distribution with zero mean and a variance of σ^2 (20). Due to the complexity and nonlinearity of the problem, the MCMC sampling technique is employed to obtain the posterior distribution of $\theta = [\mathrm{sm}, \mathrm{VOD}]$. Specifically, an ensemble MCMC sampler is implemented that iteratively explores the parametric spaces by generating multiple chains through the probabilistic transitions, converging toward the posterior distribution. The resulting MCMC chains provide posterior probabilistic predictions of SM and VOD. The maximum a posteriori (MAP) value or mode of a posterior distribution indicates the most probable value of the parameters, and the standard deviation of the posterior distribution has been used to represent retrieval uncertainty

$$\Pr\left(\theta|T_{B_p}^{\mathrm{grid}}\right) = \frac{\Pr\left(T_{B_{\mathrm{P}}}^{\mathrm{grid}}\middle|\theta\right)\Pr(\theta)}{\Pr\left(T_{B_{\mathrm{P}}}^{\mathrm{grid}}\right)} \tag{18}$$

$$\Pr\left(\theta|T_{B_{\mathrm{P}}}^{\mathrm{grid}}\right) \propto Pr\left(T_{B_{\mathrm{P}}}^{\mathrm{grid}}\middle|\theta\right)\Pr(\theta) \tag{19}$$

$$\Pr\left(T_{B_{\mathrm{P}}}^{\mathrm{grid}}\middle|\theta\right) = -\frac{1}{2}\left(\frac{\left[T_{B_{\mathrm{P}}}^{\mathrm{sim}}(\theta) - T_{B_{\mathrm{P}}}^{\mathrm{grid}}\right]^2}{\sigma^2} + \log\left(2\pi\sigma^2\right)\right). \tag{20}$$

F. Validation Metrics

Conventional evaluation metrics, including bias, unbiased root mean square error (ubRMSE), and Pearson correlation coefficient (R), have been utilized to assess the accuracy of PoLRa-derived SM retrievals, as shown from (21) to (24) [62]. In situ SM measurements from TEROS 12 probes were used as the validation benchmarks after undergoing a correction scheme. Specifically, SM estimates from the TEROS 12 probes were compared with gravimetric SM samples collected on September 13, showing a strong linear correlation (see Fig. S3). Based on this relationship, a linear correction function was derived to rescale the probe measurements to match the gravimetric SM magnitudes. This correction function was then applied to adjust probe measurements from other days, and the resulting rescaled SM values were used as benchmarks for evaluating PoLRa-derived SM retrievals. The ubRMSE quantifies discrepancies in magnitude between the retrieved and benchmarked SM, while R measures the varying consistency between the two datasets

$$\mathrm{bias} = E\left[\mathrm{sm}_{\mathrm{ret}}\right] - E\left[\mathrm{sm}_{\mathrm{ref}}\right] \tag{21}$$

$$\mathrm{RMSE} = \sqrt{E\left[\left(\mathrm{sm}_{\mathrm{ret}} - \mathrm{sm}_{\mathrm{ref}}\right)^2\right]} \tag{22}$$

$$\mathrm{ubRMSE} = \sqrt{\mathrm{RMSE}^2 - \mathrm{bias}^2} \tag{23}$$

$$R = \frac{E\left[\left(\mathrm{sm}_{\mathrm{ret}} - E\left[\mathrm{sm}_{\mathrm{ret}}\right]\right)\left(\mathrm{sm}_{\mathrm{ref}} - E\left[\mathrm{sm}_{\mathrm{ref}}\right]\right)\right]}{\sigma_{\mathrm{ret}}\sigma_{\mathrm{ref}}} \tag{24}$$

where $E[\ldots]$ represents the arithmetic mean; the subscript ret and ref denote SM retrievals from PoLRa observations and the spatially averaged in situ measurements; and σ_{ret} and σ_{ref} refer to the standard deviations of the SM retrievals and benchmarks.

IV. RESULTS AND DISCUSSION

A. Brightness Temperature Characteristics Across Land Covers and Resolutions

The collective distributions of PoLRa-observed brightness temperatures at individual footprints (T_{B_p}) have been analyzed and compared across land cover classifications and spatial resolutions. T_{B_p} data for the forested strip at 21-m resolution were excluded from Fig. 5(a) as the 7- and 14-m resolution observations did not cover this area. Using T_{B_V} data as an example, an important feature is the increased spread in

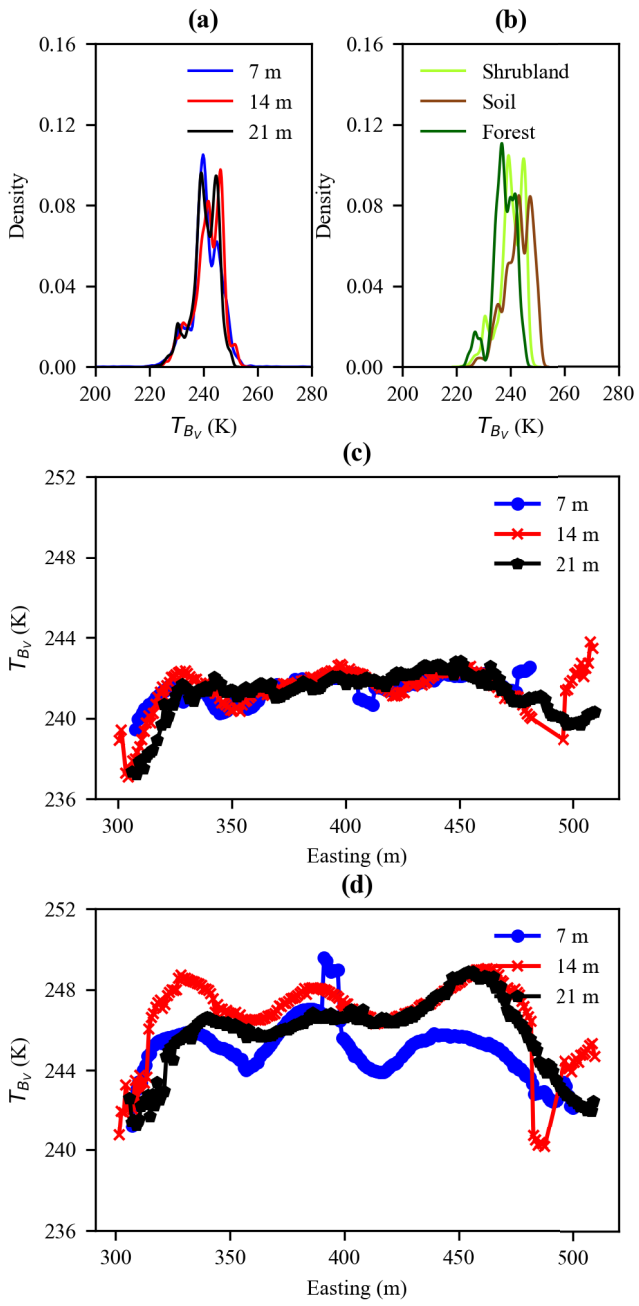


Fig. 5. (a) Probability density functions of all the vertically polarized footprint-scale brightness temperatures collected over the experiment period at the grid scales of 7, 14, and 21 m. (b) Probability density functions of 21-m vertically polarized footprint-scale brightness temperatures conditioned by land covers. (c) Variations of vertically polarized footprint-scale brightness temperatures at resolutions of 7, 14, and 21 m over Shrub 5 on September 8, 2024. (d) Variations of vertically polarized footprint-scale brightness temperatures at resolutions of 7, 14, and 21 m over bare soil on September 8, 2024.

T_{B_V} values with decreasing flight altitudes, likely reflecting the smoothing effects of coarser-resolution observations on within-pixel variability [18]. Overall, T_{B_V} distributions at multiple resolutions exhibit similar magnitudes and peak patterns around 240 K although, at medium resolution, the T_{B_V} peak shifts slightly toward higher values compared to those at low and high altitudes [see Fig. 5(a)]. In Fig. 5(b), T_{B_V} observations at a resolution of 21 m are categorized by land cover class, showing a decreasing trend in T_{B_V} values with increasing vegetation density, given that the flight paths are predominantly influenced by shrubland characteristics, leading to overlapping peaks between T_{B_V} distributions from all strips and those from the shrubland only.

Beyond aggregated distributions, footprint-scale T_{B_p} values from different resolutions have also been compared along flight tracks. Observations over Shrub 5 and Soil strips on September 8 were selected to illustrate resolution-dependent features. Given the rather dense PoLRa sampling, angular corrected T_{B_p} data were first binned every one meter along the horizontal direction, and the mean T_{B_p} values within each bin formed along-track values. Over Shrub 5, T_{B_V} values across three resolutions are highly consistent, varying by less than 2 K along the flight line [see Fig. 5(c)]. In contrast, over bare soil, larger fluctuations in T_{B_V} have been observed, which are consistently captured at all resolutions [see Fig. 5(d)]. Mid- and high-altitude T_{B_V} values are generally closer than those at low altitude, likely reflecting the closer flight time windows at 20 and 30 m. Nonetheless, the discrepancies among resolutions mostly remain within 5 K, corresponding to SM differences of approximately 2%. While the above results are solely based on T_{B_V} data, T_{B_H} observations have shown similar patterns although the gradients in T_{B_H} data across vegetation densities seem less pronounced (see Fig. S4).

These multiresolution footprint comparisons highlight expected changes across vegetation types. To further investigate spatial variability, gridded T_{B_p} maps at resolutions of 7, 14, and 21 m on September 9 have been illustrated as an instance (see Fig. 6). As expected, $T_{B_p}^{\text{grid}}$ fields become smoother with increasing flight altitude, yet key spatial features are preserved across resolutions and polarizations. $T_{B_p}^{\text{grid}}$ values are generally higher over Soil and lower over Shrub 5. Spatial patterns within shrubland strips show subtle gradients, and $T_{B_p}^{\text{grid}}$ values tend to be higher on the left side on September 9 (see Fig. 6). Comparing $T_{B_p}^{\text{grid}}$ data at medium (14 m) and low (21 m) resolutions over Shrub 2 and Shrub 3 where 7-m pixels are fully nested within larger grids shows that coarser-resolution $T_{B_p}^{\text{grid}}$ values closely approximate the spatial averages of finer-resolution observations at 7 m, consistent with previous findings [18], [63].

B. Performance Analysis of PoLRa-Based SM Estimates

Multiple algorithms based on the identical τ–ω framework have been applied to derive SM from PoLRa-observed $T_{B_p}^{\text{grid}}$ values at different altitudes. Although the magnitudes of SM retrievals vary across different algorithms, their spatial distributions generally follow the spatial patterns observed in $T_{B_p}^{\text{grid}}$ data. For example, both $T_{B_p}^{\text{grid}}$ and SM retrieval maps on September 9 display drier conditions over bare soil and a distinct east-west gradient, with drier soils in the east and wetter conditions in the west (see Fig. 7). It should be noted that the spatial patterns of SM are temporally variant. On September 4, the opposite trend, a wet east and dry west, is observed (see Fig. S5). Nevertheless, dry-to-wet SM gradients across soil, shrubland, and forest remain

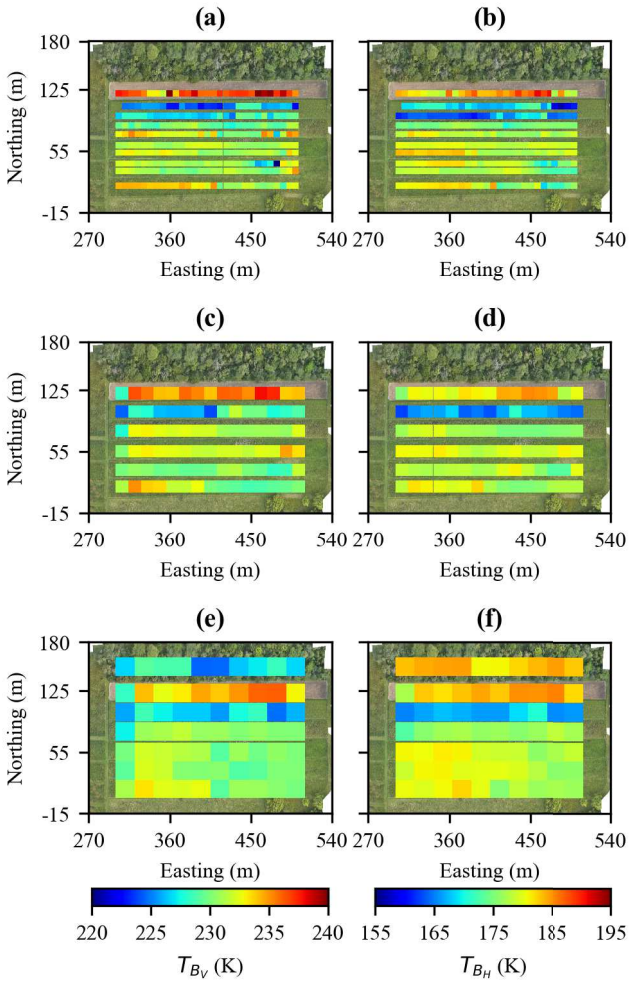


Fig. 6. Spatial maps of vertically polarized brightness temperatures at resolutions of (a) 7, (c) 14, and (e) 21 m. Spatial maps of horizontally polarized brightness temperatures at resolutions of (b) 7, (d) 14, and (f) 21 m on September 9, 2024.

consistently evident across different days. In addition, the relatively wetter tendency of Shrub 5 on September 9 has also been captured in multiple days. The reduced contrast between SM retrievals in Shrub 5 and other shrub strips, compared to the corresponding $T_{B_p}^{\text{grid}}$ patterns, may be attributed to the smoothing effects of the broad SM range and to the lower vegetation density in Shrub 5 compared to other strips. Under similar radiation intensities, denser vegetation would enhance attenuation effects and generally result in wetter SM retrievals, while sparse canopies lead to relatively drier estimates. This sensitivity is influenced by land cover type and surface conditions. Supporting this, significantly higher SCAH SM values have been obtained in the forest region than in bare soil, despite their comparable $T_{B_H}^{\text{grid}}$ values on September 9. Systematic dry DCA SM estimates on September 9 are associated with unreasonably low VOD retrievals, which could be caused by the compensation mechanism under the DCA algorithm and the original errors in $T_{B_p}^{\text{grid}}$ measurements. In contrast, DCA retrievals on September 4 produce higher SM estimates than SCA's outputs, corresponding to the elevated VOD magnitudes that mostly exceed those derived from NDVI values. When using the same VOD inputs, differences in polarized $T_{B_p}^{\text{grid}}$ measurements and Fresnel reflectivity primarily account for the differences between SCAV and SCAH SM retrievals. The MT-DCA SM estimates generally fall between those retrieved by SCA and DCA approaches (see Fig. 7).

The retrieval accuracy of PoLRa-derived SM estimates is evaluated by comparing retrievals to rescaled probe measurements using statistical metrics, including ubRMSE and R (see Table II). In contrast to RMSE, ubRMSE that is less sensitive to biases in both mean and fluctuation amplitudes has been commonly used to assess the relationship between retrievals and reference SM [10]. Independent of magnitude discrepancies, the correlation coefficient, R, measures the consistency of temporal evolutions between retrievals and reference measurements. In general, PoLRa retrievals tend to overestimate SM by approximately 0.1 m^3/m^3. Among the retrieval algorithms, ubRMSE values for SCAV and MT-DCA retrievals are generally below 0.04 m^3/m^3, meeting the SMAP mission's accuracy requirement [9].

In contrast, R values are typically lower than 0.45, partly due to the limited sample size across the nine-day experimental period, where minor anomalies could significantly affect temporal correspondence. Compared with our prior pole-fixed PoLRa study, the R values for SCA retrievals are markedly reduced, most likely because spillover effects were not fully removed, given their day-to-day variability and the imperfect derivation of the daily offset terms d_p. In addition, even before applying d_p, the temporal trends of T_{B_V} and T_{B_H} exhibit opposite tendencies on certain days, which drives negative correlations between SCAH retrievals and the in situ observations. It is inferred that these reversed trends are related to polarization-imbalanced impacts of RFI contamination, together with the use of calibration parameters that do not explicitly account for polarization dependence. For completeness, Section IV-D expands on these points and discusses, in detail, the factors that likely reduced PoLRa's ability to track SM variations temporally and produced the polarization-divergent behavior observed in this campaign. Nevertheless, the improved R values for DCA and MT-DCA retrievals suggest that the use of dual-polarization observations from the same or adjacent days can partially mitigate the effects of noisy inputs and invariant VOD across the experimental period for SCA here (see Fig. S6). Overall, MT-DCA retrievals exhibit the best integrated performance in terms of ubRMSE and R under the conditions of this field campaign.

Among different algorithms, ubRMSE values markedly increase at the 21-m resolution compared to 7 and 14 m, with the performance degradation primarily driven by poorer retrievals over the forest regions (see Fig. 8). Considering VWC distributions from Landsat 8 NDVI data and using a threshold of 0.04 m^3/m^3, PoLRa retrievals appear more accurate when VWC is below 1.5 kg/m^2 [see Fig. 8(c)]. This preference could also explain the reduced accuracy of SCA retrievals based on Sentinel-2 VOD products where mean VWC values are 1.2 kg/m^2 in shrubland and 2.5 kg/m^2 in forest (see Fig. S2). The observed degradation in ubRMSE values with increasing VWC is consistent with previous SMAP

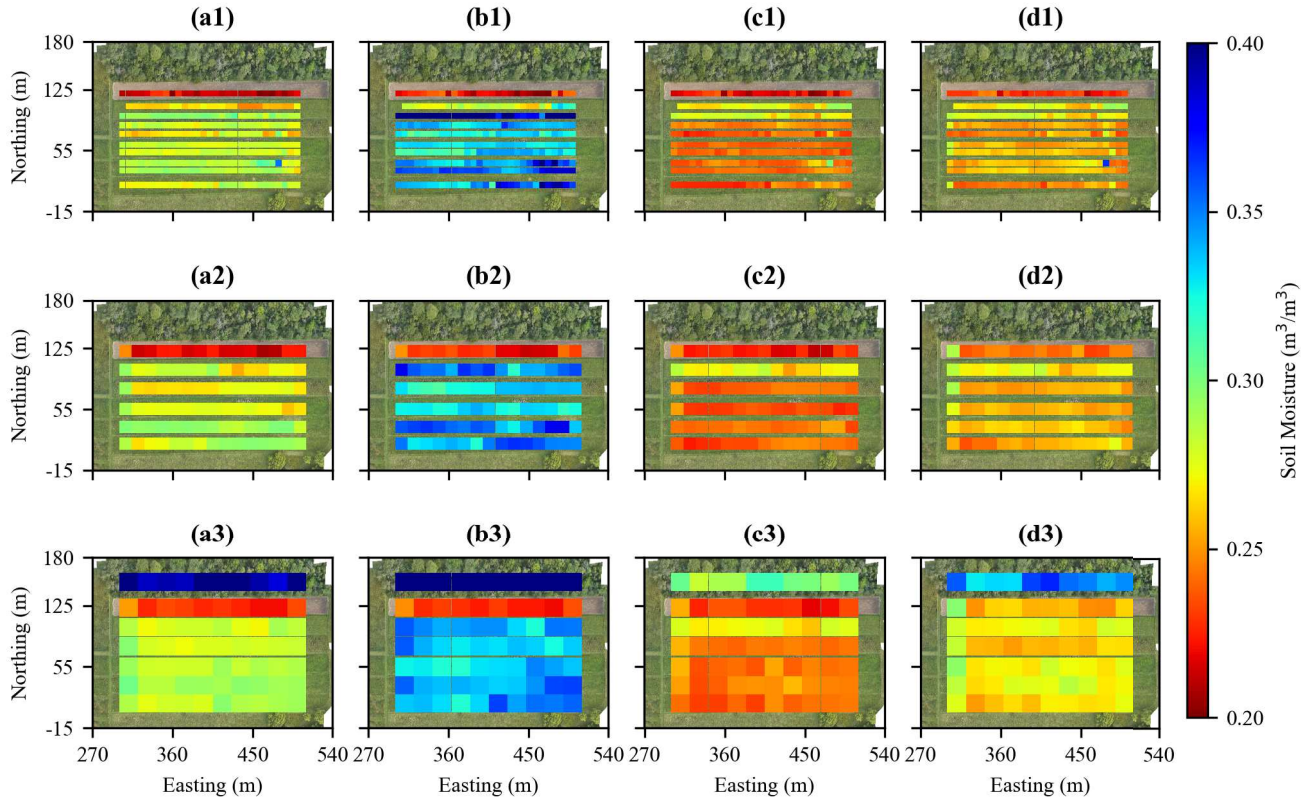


Fig. 7. Spatial distributions of SM retrievals derived from (a1) to (a3) the SCAV, (b1) to (b3) the SCAH, (c1) to (c3) the DCA, and (d1) to (d3) the MT-DCA on September 9, 2024. (Top, medium, and bottom rows) Spatial resolutions of SM at 7, 14, and 21 m, respectively.

validation studies [13]. Accurately modeling T_{B_p} data over dense canopies is a persistent challenge due to several factors that, in turn, complicate the retrieval accuracy of SM [64]. These include the presence of single and multiple scattering effects, the wide distribution in the size and orientation of forest branches, and the fact that the permittivity properties of forest branches differ significantly from those of herbaceous vegetation [64], [65], [66].

In areas with VWC below 1.5 kg/m^2, accurate representation of VOD is critical for SM retrieval precision. For MT-DCA retrievals, ubRMSE values over shrublands are approximately 0.03 m^3/m^3, slightly better than those over bare soils (see Fig. 8). In contrast, for SCAV retrievals, ubRMSE values were consistently the smallest over bare soil (~0.02 m^3/m^3) across all land cover types. Both SCAV and MT-DCA retrievals demonstrate comparable accuracy over shrubland strips although MT-DCA outperforms in Shrub 3. Examples of SM time series from PoLRa retrievals and in situ probe measurements for three representative pixels separately in Shrub 5, soil, and forest are presented in Fig. 9. Consistent with spatial patterns, SM magnitudes increase with vegetation density. Due to the absence of precipitation during the experiment, in situ probe measurements exhibit a general drying trend although minor fluctuations have also been observed. In contrast, PoLRa retrievals show two distinct drying segments: September 5–8 and September 9–12. While both the PoLRa retrievals and in situ data show an unexpected SM increase on September 9, despite zero precipitation, the PoLRa rise is most plausibly attributable to an improperly applied d_p (significantly lower—by ~20 K—than on surrounding days), which underestimates T_B. This T_B underestimation, in turn, inflates the corresponding SM retrievals.

C. Bayesian Uncertainty of PoLRa Retrievals

As mentioned earlier, Bayesian inference produces full posterior distributions of SM and VOD rather than deterministic point estimates obtained by the least-squares optimization. The MAP estimate, the mode of the posterior distribution, represents the most probable value of SM and/or VOD given the observed $T_{B_p}^{\text{grid}}$ values and prior information. Ideally, in this study, the MAP estimates and the least-squares solutions should converge as both aim to minimize the differences between modeled and observed T_B values. Such a convergence can be observed for the MAP and least-squares SM retrievals from the SCA algorithm [see Fig. 10(a) and (b)]. For the DCA-based retrieval frameworks, however, an average RMSE of approximately 0.01 m^3/m^3 indicates the existence of small discrepancies [see Fig. 10(c) and (e)]. These differences in the context of DCA-related algorithms are expected to alleviate when the bounds on SM and VOD are extended, as shown in Fig. S7. Yet, retrievals that exceed physically plausible limits could suggest: 1) significant inconsistencies between dual-polarized observations and/or 2) unrealistic $T_{B_p}^{\text{grid}}$ mea-

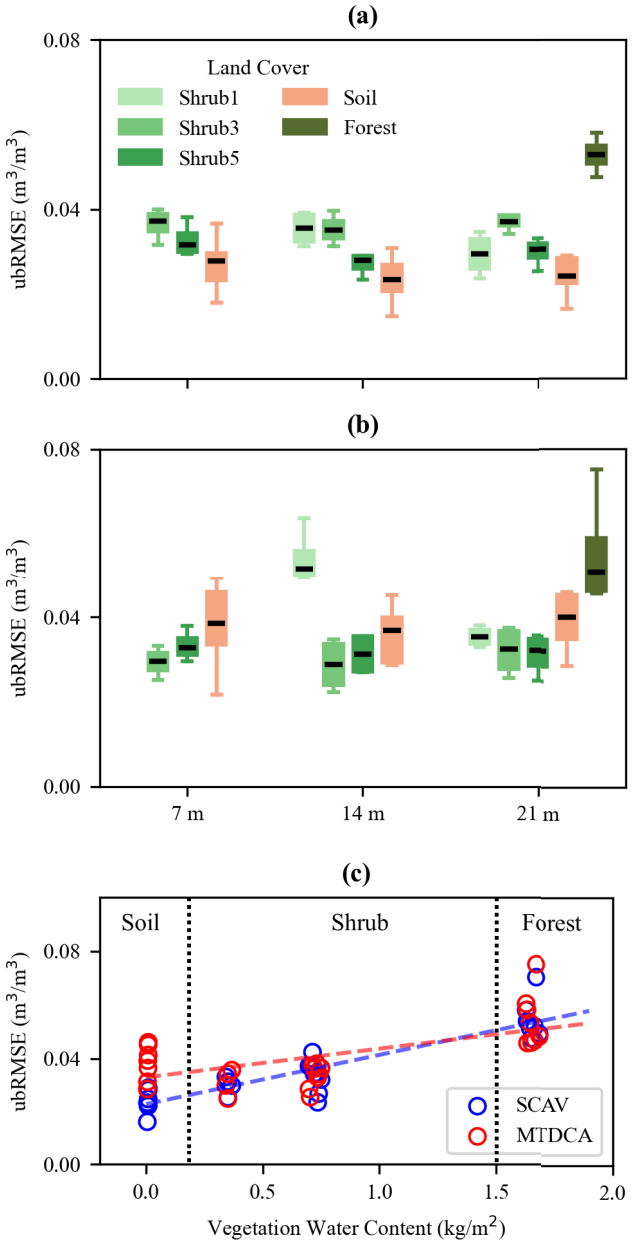


Fig. 8. Boxplots of ubRMSE values for SM retrievals derived using (a) SCAV and (b) MT-DCA across different land covers. (c) Variations of ubRMSE values for SM retrievals at the 21-m grids across gridded VWC. The vertical dashed lines are for illustrative purposes only, serving to separate the data into soil, shrub, and forest strips.

surements. Given the overall quality of SCA SM retrievals, the second explanation, unrealistic $T_{B_p}^{\text{grid}}$ measurements, is less likely, and mismatched dual-polarized $T_{B_p}^{\text{grid}}$ values induced by calibration and/or instrument issues are more plausible contributors. When optimal solutions fall outside prescribed physically reasonable bounds, least-square retrievals often drive the VOD estimate to zero, whereas MAP estimates are less prone to cluster at zero [see Fig. 10(d) and (f)]. As a result, MAP SM estimates tend to be slightly higher than their deterministic counterparts [see Fig. 10(c) and (e)]. The retrievals affected and not affected by this mechanism have been separately marked as orange and blue points shown in Fig. 10(c)–(f). Nonetheless, these differences are rather

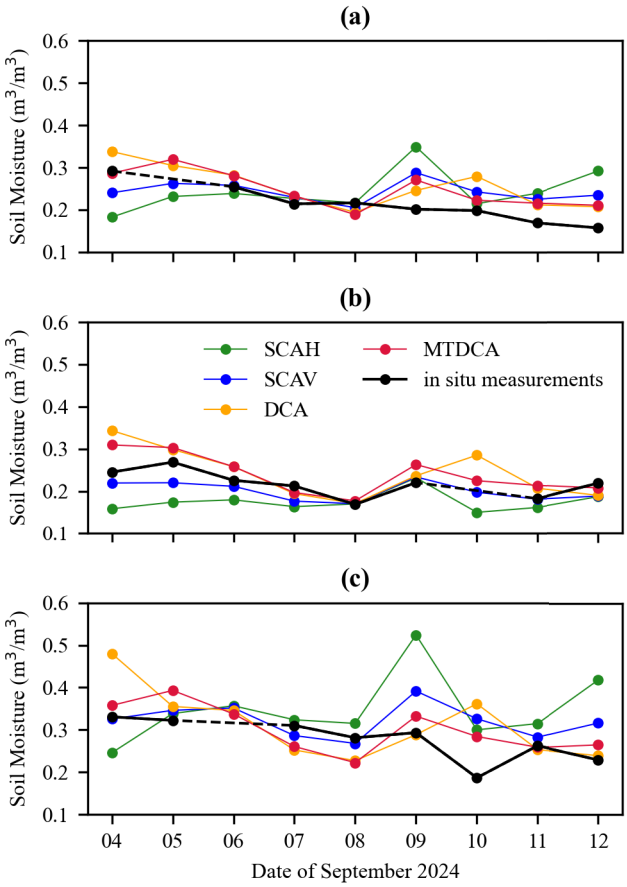


Fig. 9. Time series of in situ SM measurements (black) and PoLRa-derived retrievals using the SCAV (blue) and SCAH (green), the DCA (orange), and the MT-DCA (red) in (a) Shrub 5, (b) soil, and (c) forest. The dashed line represents that unavailable in situ data between the neighboring days.

trivial and do not necessarily degrade SM retrieval accuracy, as observational noise tends to impact VOD retrievals more severely than SM estimates [e.g., Fig. 10(f)] [67]. In practice, abrupt drops of VOD to zero over time are not uncommon for DCA-based retrievals, including the SMAP official SM product where an additional regularization term is also included in the retrieval objective function [33]. Furthermore, the full posterior distributions generated by Bayesian inference can sometimes capture cases where nonunique DCA solutions may exist. Despite these, MAP and least-square estimates generally converge closely.

The standard deviations derived from the posterior distributions have been used in this study to represent the uncertainty spreads of PoLRa-retrieved SM and VOD values. Although the posterior mean and the MAP estimates are mostly not aligned, the standard deviation surrounding the MAP estimate is typically considered equivalent to that around the mean [68]. It is important to note that in the absence of parameter uncertainty, retrievals from the SCA algorithm produce a unique deterministic solution, and the corresponding uncertainty intervals generated through Bayesian inference are extremely narrow, with an average magnitude of 0.02 m^3/m^3. This is because the error variance (σ^2) in (20), associated with a single T_B, is set to 10^{-6}, which causes the sampler

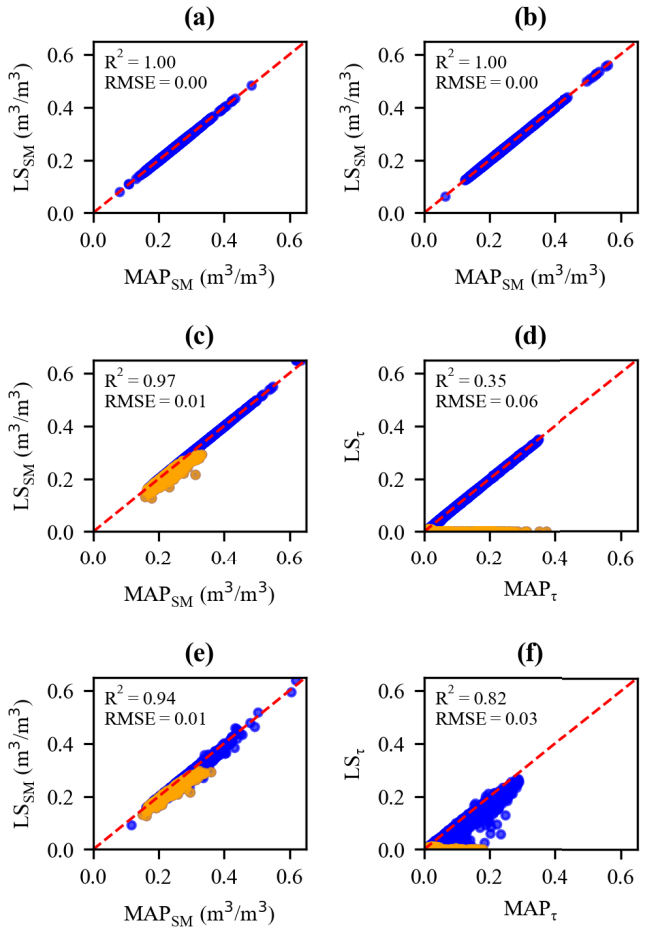


Fig. 10. Scatterplots of SM and VOD (τ) retrievals derived from the least-square optimization (LS) and the MAP based on Bayesian inference in the context of the SCA (a) at the vertical polarization and (b) horizontal polarization, (c) and (d) DCA, and (e) and (f) MT-DCA. The orange dots identify retrievals where the LS optimization has constrained τ retrievals to its physical lower bound, i.e., 0 kg/m^2. This occurs in cases where the optimal mathematical τ solutions are negative values. In contrast, the MAP estimates of VOD are not systematically forced to this zero boundary. The blue dots represent the retrievals that are optimal and fall within the physically reasonable bounds.

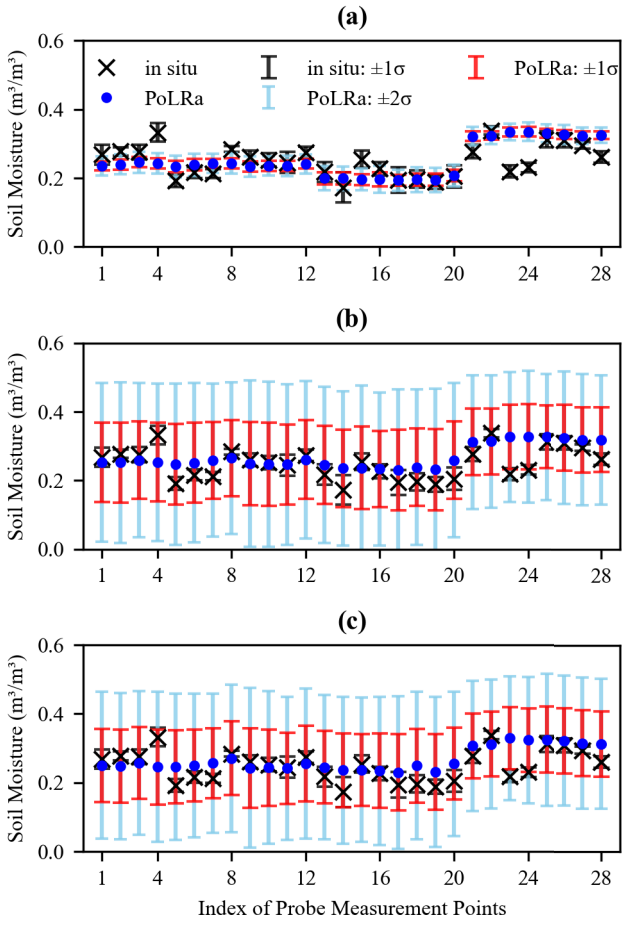


Fig. 11. Spreads of 21-m SM uncertainty are based on (a) SCA, (b) DCA, and (c) MT-DCA. In situ SM measurement uncertainty is displayed by the black bar and was estimated by averaging daily standard deviations of three SM readings over each site.

to explore a very narrow range and results in sharply peaked posterior distributions. Meanwhile, the mean standard deviations for DCA- and MT-DCA-retrieved SM are approximately ± 0.12 and ± 0.11 m^3/m^3, respectively. In DCA and MT-DCA, σ^2 is the variance of multiple residuals (e.g., both residuals of T_{B_H} and T_{B_V} in DCA). These variances change across MCMC iterations, unlike those in SCA, which remain constant and are also larger than in SCA. This broadens the likelihood, prompting the MCMC sampler to explore a wider parameter space in subsequent iterations and resulting in greater posterior uncertainty. Given this, direct comparisons of standard deviations or uncertainty ranges across various algorithms are unfair. When no perfect SM-VOD combinations that yield zero residuals for balancing polarized $T_{B_p}^{\text{grid}}$ observations exist in the predefined search spaces, a "cloud" of suboptimal solutions emerges, which significantly inflates the uncertainty range. These large residuals could be attributed to simulation-observation mismatches induced by imperfect models and improper parameters (in addition to SM and VOD) and misaligned observations across polarizations due to sensor anomaly and calibration deviations, which are beyond the scope of this work. Fig. 11 illustrates the spreads of 21-m SM uncertainty ranges based on nine-day MAP averages of different algorithms at 28 locations where probe measurements were conducted. The reference SM values largely fall within the onefold uncertainty intervals (i.e., $\pm 1\sigma$), especially over bare soil, i.e., sites 12–20 (see Fig. 11).

D. Discussion

This study expands on previous research [30] by transitioning from the stationary pole-fixed system at a few distinct sampling sites to a mobile, aerial platform for mapping high-resolution SM. The UAV-mounted PoLRa flew at three different elevations over a large field containing diverse land covers, including a forested strip. An extension of the performance assessment of UAV-radiometer systems is undertaken beyond commonly studied agricultural and bare soil sites, with greater emphasis placed on natural land cover scenarios, such as shrubland and forest. A foundation can,

thus, be laid for broader applications of UAV-mounted PoLRa across diverse environments (e.g., urban forest). In addition to land cover classes, PoLRa retrievals are evaluated across vegetation-density gradients characterized by VWC estimates from 0 to 1.8 kg/m^2. This range cannot be replicated during a short campaign at a homogeneous agricultural site where VWC is quasi-stationary. Moreover, SM retrievals are compared within nested grids, providing unique insight and indicating that UAV-mounted PoLRa can be used to explain the spatial behaviors of SM variations across scales on the order of tens of meters. Furthermore, Bayesian inference is employed to quantify retrieval uncertainty, thereby providing a more comprehensive characterization of the quality of PoLRa-based SM estimates. Beyond facilitating its applications, these investigations are also considered crucial for improving the accuracy of UAV-based PoLRa SM retrievals by identifying current limitations and delineating directions for future enhancement.

Compared with the high retrieval accuracy achieved by the pole-fixed PoLRa configuration, the UAV-based PoLRa SM retrievals exhibit broadly similar performance in terms of magnitude estimation. However, a moderate decline in the temporal consistency between the UAV-based SM retrievals and the in situ probe measurements has been observed. This degradation is attributable primarily to unexpected radio frequency interference (RFI) contamination and the imperfect application of the daily offset (i.e., d_p). Prior to applying these d_p values, the cold-sky T_{B_p} recorded throughout the campaign predominantly fall within 60–70 K, substantially higher than the expected value of approximately 5 K for an uncontaminated cold-sky view [44] (see Fig S8). This elevated signal drives the corresponding SM retrievals toward 0 m^3/m^3 and strongly indicates the presence of RFI-induced spillovers. Although the brief sky measurements were originally designed as daily sanity checks rather than as inputs to a full parameter calibration, the anomalously high T_{B_p} values necessitated repurposing the cold-sky readings to estimate d_p and, thereby, offset spillover effects, bringing T_{B_p} back into a physically plausible range so that the analysis could proceed. However, day-to-day variability in RFI and subtle differences in the sky field-of-view (e.g., the geometric setups) render these cold-sky snapshots inadequate for a robust calibration procedure. Moreover, applying d_p does not guarantee complete removal of daily RFI perturbations that influence the temporal evolution of the surface-emission signal. Consequently, residual RFI contamination can depress the temporal correspondence between the retrievals and the in situ measurements (even after offsetting). Furthermore, we identified a likely misestimated d_p on September 9, which produces a sudden, artificial increase in the SM time series on that date (see Fig. 9). Notably, this RFI-induced spillover behavior was neither anticipated nor observed in our prior pole-fixed experiment, further supporting the conclusion that episodic RFI contamination, rather than platform changes, underlies the degraded temporal fidelity observed in this study.

Given the strong linear relation between T_{B_p} and their derived SM retrievals implied by the τ–ω model, examining the temporal variability of T_{B_H} and T_{B_V} could provide useful clues for understanding the negative correlations observed between the SCAH SM retrievals and probe measurements. Prior to applying d_p values, the overall temporal correlation between the T_{B_H} and T_{B_V} time series remains negative (approximately −0.5), a result that stands in contrast to the strong positive correlation evident in their voltage readings (see Fig S9). However, this apparent inconsistency can arise when $(u_p - u_{\mathrm{RS}})$ differs across polarizations, given the formulation $T_{B_p} = G \cdot (u_p - u_{\mathrm{RS}}) + T_{B_{\mathrm{RS}}}$ derived from (1) to (3). As an illustrative example, during the final two observation days characterized by nearly identical ambient temperatures, the temporal tendencies of $(u_{\mathrm{RS}} - u_V)$ and $(u_{\mathrm{RS}} - u_H)$ are opposite (see Fig S10). This uneven correction is then amplified by the multiplication by G, which propagates into polarization-specific divergences in the retrieved SM: SCAV decreases, while SCAH increases on September 11 and 12 [see Fig. 9(a) and (b)]. We infer that the initial voltage imbalance between the H- and V-polarized channels likely reflects the polarized character of RFI contamination and an incomplete consideration of polarization effects within the adopted calibration parameter set. While the employment of d_p values can somewhat mitigate their divergence, it cannot be guaranteed that the polarized-dependent differences are minimized in a manner that preserves the true temporal evolution of the surface emissions.

Beyond the disruption caused by the unexpected signal spillovers and the imperfect use of d_p, discrepancies between PoLRa retrievals and in situ SM benchmarks were also partly attributable to an inadequate treatment of instrument thermal drift. Reliance on brief cold-sky snapshots together with ambient temperature for calibration can introduce temperature-related bias. During the campaign, the ambient air temperatures spanned approximately from 16 °C to 28 °C. Our analysis did not reveal a statistically significant relationship between the daily temperature and the measured cold sky T_{B_p} (see Fig. S8). The onboard ACS and RS typically maintain radiation intensity of roughly 20 K and ~310 K, bracketing the expected land-surface T_{B_p} ranges and supporting two-point interpolation rather than extrapolation. In addition, the prior work indicates that, for PoLRa, a single parameter set derived from the ground-based temperature-variant sky calibration can remain valid across a reasonably broad range of thermal conditions. These features could lend practical confidence that PoLRa retains applicability with limited interference to ambient temperature in the absence of a hot external reference. However, in this study, the superposition of episodic RFI contamination and day-to-day variations in the geometric configuration of the cold-sky measurements prevents a clean attribution of the low correlation between T_{B_p} and ambient temperature. In other words, the weak correlation cannot be interpreted as evidence that thermal drift effects are fully absent. Given these limitations, a hot calibration source, such as a microwave absorber with known T_{B_p} values, will also be required to mitigate temperature-driven bias at warm conditions [18]. Furthermore, inclusion of an in-flight reference target (e.g., water bodies) would also be beneficial for improving calibration by including flight-dynamic perturbations. In addition, factors such as the

temperature gradients within the instrument and the unwanted signals from solar intrusion would also be potential error sources.

Furthermore, inadequate spatial representativeness of point measurements could also be a key contributor to the discrepancies. For example, probe measurements in forested strips were located mostly at the edges of the radiometer footprints rather than at multiple, spatially representative points within the footprint. To maximize the number of evaluation scenarios, some in situ measurements from outside the pixels were still used as benchmarks for their most adjacent grids. These setups likely biased the validation metrics, resulting in an underestimation of retrieval performance. Furthermore, the short ten-day period renders the correlation metrics exceptionally sensitive to minor fluctuations.

On the algorithmic side, the unavailability of fine-scale model parameters (e.g., h and ω) complicates the retrieval of high-precision SM estimates. Land cover-dependent parameters have been adopted for satellite missions such as SMAP [9]. Although these empirical values have initially been derived from field campaigns, recent studies have shown that ω can vary more within a given land cover class than between different classes [38]. Furthermore, h and ω parameters are known to vary with landscape conditions, temporal dynamics, and algorithm type [69], [70]. While SM retrievals based on h of 0.13 and ω of 0.08 have exhibited small ubRMSE values, these choices are likely suboptimal and cannot generalize across all land covers. To mitigate this concern, we ran a sensitivity analysis to assess how combinations of h and ω affect retrieval accuracy. We computed the ubRMSE for SM estimates across h from 0.1 to 0.2 and ω from 0 to 0.1, fully covering the SMAP ancillary parameter space [9]. The ubRMSE varies only slightly within an extent of 0.01 m^3/m^3 for all tested pairs, indicating that, within the context of this study, using a suboptimal h has a limited impact on the final SM retrievals (see Fig. S11). Another critical parameter for SCA-based retrievals is the b_{LC} coefficient, which translates VWC into VOD and plays a key role during active vegetation growth periods [17]. In addition, without on-site validation, the NDVI-derived VOD may be ill-suited for this setting and can adversely affect the accuracy of SCA-based retrievals. Moreover, this study does not account for vertical temperature gradients within the vegetation layer or for deeper soil temperatures when estimating effective temperature, which can bias the partitioning of emission and consequently introduce error into the SM retrievals. Intracampaign temperature variability over the ~2-h field windows can alter effective temperatures and degrade retrieval quality.

Despite the abovementioned temporal limitations, the instrument's daily spatial performance remains strong. On most days, the spatial correlations between $T_{B_p}^{\text{in}}$ and u_p generally exceed 0.7. In addition, the spatial correspondence between gridded $T_{B_p}^{\text{in}}$ and in situ SM measurements consistently shows the physically expected negative correlation. The pronounced SM gradients across strips with different vegetation densities provide further corroborating evidence of spatial sensitivity. Taken together, these results suggest that, while temporal correlation performance is vulnerable to daily variable RFI contamination, the PoLRa retrievals are still reliable for capturing spatial SM gradients at the scales examined.

Regarding SM retrieval uncertainty, the feasibility of applying Bayesian inference to drone-based PoLRa observations has been examined using τ–ω-based algorithms. Bayesian inference provides probabilistic retrievals instead of single-value deterministic estimates—enabling determination of the confidence or uncertainty of the retrievals. It also provides additional insights by depicting the near-optimal solutions in the posterior distributions of the retrieved parameters. Following this, the current framework could be extended by incorporating prior distributions for model parameters with both SCA and DCA schemes. However, it should be noted that fully simultaneous retrievals of optimal parameter values and their associated uncertainties would require richer observational information. For drone-based systems, this could potentially be achieved by adjusting flight configurations to collect multiangular observations over short windows. It is also important to note that MAP estimates are effective parameters optimized for retrieval purposes and may not always reflect their true physical meanings [64].

V. Conclusion

This study conducted a comprehensive evaluation of SM retrievals using the PoLRa mounted on a UAV platform across diverse land covers and spatial resolutions. A series of controlled flights was performed over shrubland, bare soil, and forest strips, with data collected at multiple altitudes to generate gridded brightness temperatures ($T_{B_p}^{\text{grid}}$) at spatial scales of 7, 14, and 21 m. Footprint-scale and gridded (polarized) brightness temperatures (T_{B_p}) were analyzed to assess resolution-dependent variability and investigate the impacts of vegetation density on microwave emissions. The results reveal that, although coarser resolutions tend to smooth spatial variations, key land surface features were preserved across resolutions. Notably, $T_{B_p}^{\text{grid}}$ distributions show consistent dry–wet gradients associated with changes in vegetation density, and spatial correlations between vertical and horizontal polarizations were generally maintained across scales.

Building on the $T_{B_p}^{\text{grid}}$ observations, SM retrievals were carried out using multiple τ–ω-based algorithms, including the SCA, DCA, and its derivative MT-DCA. SM retrievals from the SCAV and MT-DCA algorithms achieve ubRMSEs generally below 0.04 m^3/m^3, meeting the SMAP mission's target accuracy requirement. Among all retrieval methods, MT-DCA consistently produces the best overall performance, with ubRMSE values of around 0.03 m^3/m^3 and correlation coefficient (R) values of around 0.45 over shrubland and bare soils. On the other hand, SCAH retrievals suffer from low or even negative R values, possibly highlighting calibration issues, particularly at horizontal polarization. Compared to the DCA and SCAH, MT-DCA retrievals exhibit substantially improved robustness, demonstrating greater tolerance to measurement noise, calibration uncertainties, and imperfect model assumptions through the exploitation of multitemporal observation constraints. SM retrieval accuracy of PoLRa is generally higher over areas with VWC less than 1.5 kg/m^2,

and retrieval performance degrades over forested regions with denser vegetation. Bayesian inference results confirmed that reference SM measurements largely fall within the onefold derived uncertainty intervals surrounding the maximum a posteriori (MAP) estimates. While the standard deviations surrounding the SCAV and SCAH MAP estimates approximate ± 0.02 m^3/m^3, the uncertainty ranges for DCA and MT-DCA are ± 0.12 and ± 0.11 m^3/m^3, potentially due to model-observation discrepancies and/or misaligned observations across polarizations. These findings highlight the strong potential of UAV-based PoLRa for high and scalable resolution SM sensing across multiple land cover conditions with continuous refinement on calibration techniques and standardization.

ACKNOWLEDGMENT

The authors gratefully acknowledge James L. Ellis and Nathan C. Hudson for their support in providing and managing the experimental site. They also thank Daniel Jimenez Gil and Shamontee Aziz for their assistance with field data collection and soil analysis.

REFERENCES

[1] D. Entekhabi et al., "The soil moisture active passive (SMAP) mission," *Proc. IEEE*, vol. 98, no. 5, pp. 704–716, May 2010.

[2] R. D. Koster et al., "Regions of strong coupling between soil moisture and precipitation," *Science*, vol. 305, no. 5687, pp. 1138–1140, Aug. 2004.

[3] G. Petropoulos, *Remote Sensing of Energy Fluxes and Soil Moisture Content*. Boca Raton, FL, USA: CRC Press, 2013.

[4] S. I. Seneviratne et al., "Investigating soil moisture–climate interactions in a changing climate: A review," *Earth-Sci. Rev.*, vol. 99, nos. 3–4, pp. 125–161, 2010.

[5] L. Brocca, L. Ciabatta, C. Massari, S. Camici, and A. Tarpanelli, "Soil moisture for hydrological applications: Open questions and new opportunities," *Water*, vol. 9, no. 2, p. 140, Feb. 2017.

[6] B. Fang, V. Lakshmi, R. Bindlish, T. J. Jackson, and P.-W. Liu, "Evaluation and validation of a high spatial resolution satellite soil moisture product over the continental United States," *J. Hydrol.*, vol. 588, Sep. 2020, Art. no. 125043.

[7] E. Jalilvand, R. Abolafia-Rosenzweig, M. Tajrishy, and N. N. Das, "Evaluation of SMAP/Sentinel 1 high-resolution soil moisture data to detect irrigation over agricultural domain," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 10733–10747, 2021.

[8] P. K. Krishnamurthy R, J. B. Fisher, R. J. Choularton, and P. M. Kareiva, "Anticipating drought-related food security changes," *Nature Sustainability*, vol. 5, no. 11, pp. 956–964, Sep. 2022.

[9] P. O'Neill et al., "Algorithm theoretical basis document level 2 & 3 soil moisture (passive) data products," *SMAP Project*, Jet Propulsion Laboratory, Pasadena, CA, USA, Tech. Rep. JPL D-66480, Oct. 2021. Accessed: Mar. 18, 2024. [Online]. Available: https://nsidc.org/sites/nsidc.org/files/technical-references/L2_SM_P_ATBD_rev_G_final_Oct2021.pdf

[10] S. Chan et al., "Development and assessment of the SMAP enhanced passive soil moisture product," *Remote Sens. Environ.*, vol. 204, pp. 931–941, Jan. 2017.

[11] S. K. Chan et al., "Assessment of the SMAP passive soil moisture product," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4994–5007, Aug. 2016.

[12] A. Colliander et al., "Validation of SMAP surface soil moisture products with core validation sites," *Remote Sens. Environ.*, vol. 191, pp. 215–231, Mar. 2017.

[13] R. Zhang, S. Kim, and A. Sharma, "A comprehensive validation of the SMAP enhanced level-3 soil moisture product using ground measurements over varied climates and landscapes," *Remote Sens. Environ.*, vol. 223, pp. 82–94, Mar. 2019.

[14] R. Zhang, S. Kim, A. Sharma, and V. Lakshmi, "Identifying relative strengths of SMAP, SMOS-IC, and ASCAT to capture temporal variability," *Remote Sens. Environ.*, vol. 252, Jan. 2021, Art. no. 112126.

[15] J. Peng et al., "A roadmap for high-resolution satellite soil moisture applications—Confronting product characteristics with user requirements," *Remote Sens. Environ.*, vol. 252, Jan. 2021, Art. no. 112162.

[16] A. Colliander et al., "Comparison of high-resolution airborne soil moisture retrievals to SMAP soil moisture during the SMAP validation experiment 2016 (SMAPVEX16)," *Remote Sens. Environ.*, vol. 227, pp. 137–150, Jun. 2019.

[17] A. Colliander et al., "Retrieving soil moisture for non-forested areas using PALS radiometer measurements in SMAPVEX12 field campaign," *Remote Sens. Environ.*, vol. 184, pp. 86–100, Oct. 2016.

[18] T. J. Jackson, "Multiple resolution analysis of L-band brightness temperature for soil moisture," *IEEE Trans. Geosci. Remote Sens.*, vol. 39, no. 1, pp. 151–164, Jan. 2001.

[19] T. N. Carlson, "The right triangle model: Overcoming the sparse data problem in thermal/optical remote sensing of soil moisture," *Remote Sens.*, vol. 16, no. 17, p. 3231, Aug. 2024.

[20] B. Fang, V. Lakshmi, M. H. Cosh, and C. Hain, "Very high spatial resolution downscaled SMAP radiometer soil moisture in the CONUS using VIIRS/MODIS data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 4946–4965, 2021.

[21] B. Fang, V. Lakshmi, and R. Zhang, "Validation of downscaled 1-km SMOS and SMAP soil moisture data in 2010–2021," *Vadose Zone J.*, vol. 23, no. 2, Mar. 2024, Art. no. e20305.

[22] A. Balenzano et al., "Sentinel-1 soil moisture at 1 km resolution: A validation study," *Remote Sens. Environ.*, vol. 263, Sep. 2021, Art. no. 112554.

[23] E. Dai, A. J. Gasiewski, A. Venkitasubramony, M. Stachura, and J. Elston, "High spatial resolution soil moisture mapping using a lobe differencing correlation radiometer on a small unmanned aerial system," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4062–4079, May 2021.

[24] N. Ye, J. P. Walker, Y. Gao, I. PopStefanija, and J. Hills, "Comparison between thermal-optical and L-band passive microwave soil moisture remote sensing at farm scales: Towards UAV-based near-surface soil moisture mapping," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 633–642, 2024.

[25] K. Y. Kim et al., "Precision soil moisture monitoring with passive microwave L-band UAS mapping," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 7684–7694, 2024.

[26] D. Houtz, R. Naderpour, and M. Schwank, "Portable L-band radiometer (PoLRa): Design and characterization," *Remote Sens.*, vol. 12, no. 17, p. 2780, Aug. 2020.

[27] E. M. McIntyre and A. J. Gasiewski, "An ultra-lightweight L-band digital lobe-differencing correlation radiometer (LDCR) for airborne UAV SSS mapping," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2007, pp. 1095–1097.

[28] R. Acevo-Herrera et al., "Design and first results of an UAV-borne L-band radiometer for multiple monitoring purposes," *Remote Sens.*, vol. 2, no. 7, pp. 1662–1679, Jun. 2010.

[29] S. Lv et al., "Rapid retrieval of soil moisture using a novel portable L-band radiometer in the Hulunbeier Prairie, China," *GISci. Remote Sens.*, vol. 61, no. 1, Dec. 2024, Art. no. 2424337.

[30] R. Zhang, A. Nayak, D. Houtz, A. Watts, E. Soltanaghai, and M. Alipour, "Evaluation of soil moisture retrievals from a portable L-band microwave radiometer," *Remote Sens.*, vol. 16, no. 23, p. 4596, Dec. 2024.

[31] M. Sapkota, C. M. Straw, W. W. Floyd, and E. Scudiero, "Portable L-band radiometry for soil moisture mapping on golf course fairways: A calibration study with analysis of covariance regression in Texas," *Int. Turfgrass Soc. Res. J.*, pp. 1–7, Mar. 2025, doi: 10.1002/its2.70018.

[32] (2025). *DJI Mavic 3E/3T User Manual V1.0*. Accessed: May 26, 2025. [Online]. Available: https://dl.djicdn.com/downloads/DJI_Mavic_3_Enterprise/DJI_Mavic_3E_3T_User_Manual_EN.pdf

[33] J. Chaubell et al., "Regularized dual-channel algorithm for the retrieval of soil moisture and vegetation optical depth from SMAP measurements," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 102–114, 2022.

[34] L. Gao, M. Sadeghi, and A. Ebtehaj, "Microwave retrievals of soil moisture and vegetation optical depth with improved resolution using a combined constrained inversion algorithm: Application for SMAP satellite," *Remote Sens. Environ.*, vol. 239, Mar. 2020, Art. no. 111662.

[35] A. G. Konings, D. Entekhabi, S. K. Chan, and E. G. Njoku, "Effect of radiative transfer uncertainty on L-band radiometric soil moisture retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 7, pp. 2686–2698, Jul. 2011.

[36] A. G. Konings, K. A. McColl, M. Piles, and D. Entekhabi, "How many parameters can be maximally estimated from a set of measurements?," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 5, pp. 1081–1085, May 2015.
[37] A. G. Konings, M. Piles, N. Das, and D. Entekhabi, "L-band vegetation optical depth and effective scattering albedo estimation from SMAP," *Remote Sens. Environ.*, vol. 198, pp. 460–470, Sep. 2017.
[38] A. G. Konings, M. Piles, K. Rötzer, K. A. McColl, S. K. Chan, and D. Entekhabi, "Vegetation optical depth and scattering albedo retrieval using time series of dual-polarized L-band radiometer observations," *Remote Sens. Environ.*, vol. 172, pp. 178–189, Jan. 2016.
[39] A. Lye, A. Cicirello, and E. Patelli, "Sampling methods for solving Bayesian model updating problems: A tutorial," *Mech. Syst. Signal Process.*, vol. 159, Oct. 2021, Art. no. 107760.
[40] I. Aziz, E. Soltanaghai, A. Watts, and M. Alipour, "Bayesian inversion of GPR waveforms for sub-surface material characterization: An uncertainty-aware retrieval of soil moisture and overlaying biomass properties," *Remote Sens. Environ.*, vol. 313, Nov. 2024, Art. no. 114351.
[41] A. Fragkos, D. Loukatos, G. Kargas, and K. G. Arvanitis, "Response of the TEROS 12 soil moisture sensor under different soils and variable electrical conductivity," *Sensors*, vol. 24, no. 7, p. 2206, Mar. 2024.
[42] E. L. de la Jarrige, L. Escotte, J. M. Goutoule, E. Gonneau, and J. Rayssac, "SiGe HBT-based active cold load for radiometer calibration," *IEEE Microw. Wireless Compon. Lett.*, vol. 20, no. 4, pp. 238–240, Apr. 2010.
[43] S. S. Sobjaerg, N. Skou, and J. E. Balling, "Measurements on active cold loads for radiometer calibration," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 9, pp. 3134–3139, Sep. 2009.
[44] T. Pellarin et al., "Two-year global simulation of L-band brightness temperatures over land," *IEEE Trans. Geosci. Remote Sens.*, vol. 41, no. 9, pp. 2135–2139, Sep. 2003.
[45] G. J. M. De Lannoy et al., "Converting between SMOS and SMAP level-1 brightness temperature observations over nonfrozen land," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 9, pp. 1908–1912, Sep. 2015.
[46] S. Chan, E. Njoku, and A. Colliander, "Soil moisture active passve (SMAP): Algorithm theoretical basis document—Level 1C radiometer data product, revision A," Jet Propuls. Lab., California Inst. Technol., Pasadena, CA, USA, Tech. Rep., 2014.
[47] D. Shepard, "A two-dimensional interpolation function for irregularly-spaced data," in *Proc. 23rd ACM Nat. Conf.*, 1968, pp. 517–524.
[48] T. Mo, B. J. Choudhury, T. J. Schmugge, J. R. Wang, and T. J. Jackson, "A model for microwave emission from vegetation-covered fields," *J. Geophys. Res., Oceans*, vol. 87, no. C13, pp. 11229–11237, Dec. 1982.
[49] V. L. Mironov, L. G. Kosolapova, and S. V. Fomin, "Physically and mineralogically based spectroscopic dielectric model for moist soils," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 7, pp. 2059–2070, Jul. 2009.
[50] J. R. Wang and B. J. Choudhury, "Remote sensing of soil moisture content, over bare field at 1.4 GHz frequency," *J. Geophys. Res., Oceans*, vol. 86, no. C6, pp. 5277–5282, Jun. 1981.
[51] T. J. Jackson and T. J. Schmugge, "Vegetation effects on the microwave emission of soils," *Remote Sens. Environ.*, vol. 36, no. 3, pp. 203–212, Jun. 1991.
[52] M. Zhao, V. Humphrey, A. F. Feldman, and A. G. Konings, "Temperature is likely an important omission in interpreting vegetation optical depth," *Geophys. Res. Lett.*, vol. 51, no. 15, p. 2024, Aug. 2024.
[53] W. T. Crow et al., "An observing system simulation experiment for hydros radiometer-only soil moisture products," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 6, pp. 1289–1303, Jun. 2005.
[54] E. Fagerlund, B. Kleman, L. Sellin, and H. Svensson, "Physical studies of nature by thermal mapping," *Earth-Sci. Rev.*, vol. 6, no. 3, pp. 169–180, Jun. 1970.
[55] S. R. J. Ramson et al., "A self-powered, real-time, LoRaWAN IoT-based soil health monitoring system," *IEEE Internet Things J.*, vol. 8, no. 11, pp. 9278–9293, Jun. 2021.
[56] K. Saleh, J.-P. Wigneron, P. de Rosnay, J.-C. Calvet, and Y. Kerr, "Semi-empirical regressions at L-band applied to surface soil moisture retrievals over grass," *Remote Sens. Environ.*, vol. 101, no. 3, pp. 415–426, Apr. 2006.
[57] W. T. Crow and A. F. Feldman, "Vegetation signal crosstalk present in official SMAP surface soil moisture retrievals," *Remote Sens. Environ.*, vol. 316, Jan. 2025, Art. no. 114466.
[58] S. Feng et al., "Can real-time NDVI observations better constrain SMAP soil moisture retrievals?," *Remote Sens. Environ.*, vol. 318, Mar. 2025, Art. no. 114569, doi: 10.1016/j.rse.2024.114569.
[59] M. Drusch et al., "Sentinel-2: ESA's optical high-resolution mission for GMES operational services," *Remote Sens. Environ.*, vol. 120, pp. 25–36, May 2012.
[60] D. P. Roy et al., "Landsat-8: Science and product vision for terrestrial global change research," *Remote Sens. Environ.*, vol. 145, pp. 154–172, Apr. 2014.
[61] J. L. Beck and L. S. Katafygiotis, "Updating models and their Uncertainties. I: Bayesian statistical framework," *J. Eng. Mech.*, vol. 124, no. 4, pp. 455–461, Apr. 1998.
[62] D. Entekhabi, R. H. Reichle, R. D. Koster, and W. T. Crow, "Performance metrics for soil moisture retrievals and application requirements," *J. Hydrometeorol.*, vol. 11, no. 3, pp. 832–840, Jun. 2010.
[63] M. Drusch, E. F. Wood, and C. Simmer, "Up-scaling effects in passive microwave remote sensing: ESTAR 1.4 GHz measurements during SGP '97," *Geophys. Res. Lett.*, vol. 26, no. 7, pp. 879–882, Apr. 1999.
[64] J.-P. Wigneron et al., "Modelling the passive microwave signature from land surfaces: A review of recent results and application to the L-band SMOS & SMAP soil moisture retrieval algorithms," *Remote Sens. Environ.*, vol. 192, pp. 238–262, Apr. 2017.
[65] M. Kurum, R. H. Lang, P. E. O'Neill, A. T. Joseph, T. J. Jackson, and M. H. Cosh, "A first-order radiative transfer model for microwave radiometry of forest canopies at L-band," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 9, pp. 3167–3179, Sep. 2011.
[66] P. Ferrazzoli, L. Guerriero, and J.-P. Wigneron, "Simulating L-band emission of forests in view of future satellite applications," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 12, pp. 2700–2708, Dec. 2002.
[67] A. F. Feldman, D. Chaparro, and D. Entekhabi, "Error propagation in microwave soil moisture and vegetation optical depth retrievals," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 11311–11323, 2021.
[68] G. J. M. De Lannoy, R. H. Reichle, and J. A. Vrugt, "Uncertainty quantification of GEOS-5 L-band radiative transfer model parameters using Bayesian inference and SMOS observations," *Remote Sens. Environ.*, vol. 148, pp. 146–157, May 2014.
[69] V. A. Walker, V. Wallace, E. Yildirim, W. E. Eichinger, M. H. Cosh, and B. K. Hornbuckle, "From field observations to temporally dynamic soil surface roughness retrievals in the U.S. corn belt," *Remote Sens. Environ.*, vol. 287, Mar. 2023, Art. no. 113458.
[70] R. Zhang, A. C. Watts, and M. Alipour, "Inverse dynamic parameter identification for remote sensing of soil moisture from SMAP satellite observations," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 16592–16607, 2024, doi: 10.1109/JSTARS.2024.3457941.



Runze Zhang received the B.S. degree in chemistry and the B.A. degree in law (dual degree) from China Agricultural University, Beijing, China, in 2014, the M.S. degree in environmental engineering from the University of New South Wales, Sydney, NSW, Australia, in 2018, and the Ph.D. degree in civil engineering from the University of Virginia, Charlottesville, VA, USA, in 2023.

He is currently a Post-Doctoral Researcher with the Department of Civil and Environmental Engineering, University of Illinois Urbana–Champaign, Urbana, IL, USA. His research interests encompass the application of microwave remote sensing data on land hydrology, land surface emission modeling, and the integration of unmanned aerial vehicle sensor data to investigate surface responses to wildfires.



Ishfaq Aziz received the B.Sc. and M.Sc. degrees in civil engineering from Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh, in 2017 and 2020, respectively. He is currently pursuing the Ph.D. degree in civil and environmental Engineering with the University of Illinois Urbana–Champaign (UIUC), Urbana, IL, USA.

His research interest lies in radar sensing and nondestructive evaluation (NDE) of subsurface materials, forward modeling of radar wave propagation, and inverse parameter estimation using Bayesian Inference and physics-informed machine learning (PIML).

Derek A. Houtz (Member, IEEE) was born in Washington, DC, USA, in September 5, 1989. He received the B.S. and Ph.D. degrees in aerospace engineering from the University of Colorado, Boulder, CO, USA, in 2011 and 2017, respectively.

He currently works as a Research Scientist at the Microwave Remote Sensing group of the Swiss Federal Institute for Forest, Snow, and Landscape Research (WSL), Birmensdorf, Switzerland. He is also the Co-Founder and the C.E.O. of TerraRad Tech AG, Zurich, Switzerland. He previously worked as a Post-Doctoral Researcher at the National Institute of Standards and Technology, Boulder. His research interests include radiometer calibration, radio frequency electronics, remote sensing of the cryosphere, turfgrass soil moisture, precision irrigation, terrestrial radiometry, and radar applications.



Yuxiang Zhao received the B.S. degree in civil engineering from Tongji University, Shanghai, China, in 2019, and the M.S. degree in civil engineering from the University of Michigan, Ann Arbor, MI, USA, in 2020. He is currently pursuing the Ph.D. degree in civil engineering with the University of Illinois Urbana–Champaign, Champaign, IL, USA.

His research interests include applying robotics, computer vision, and deep learning to infrastructure inspection.



Trent W. Ford received the bachelor's degree in geography from Illinois State University, Normal, IL, USA, in 2011, and the M.S. and Ph.D. degrees from Texas A&M University, College Station, TX, USA, in 2013 and 2015, respectively.

He is Illinois State Climatologist and has been with the Prairie Research Institute, University of Illinois, Champaign, IL, USA, since 2019. He is an expert in climate and climate change and its interactions and impacts, especially as they relate to water. He leads climate and drought monitoring, research, and data collection for the state of Illinois. His team works with communities, industries, and state agencies on issues related to weather, climate, and climate change in Illinois and engages the general public through online, media, and educational outreach.



Adam C. Watts received the B.S. degree in biology and human and natural ecology (co-major) from Emory University, Atlanta, GA, USA, in 1999, and the M.S. and Ph.D. degrees in interdisciplinary ecology from the University of Florida, Gainesville, FL, USA, in 2002 and 2012, respectively.

He was a Post-Doctoral Research Ecologist with the School of Forest, Fisheries, and Geomatics Sciences, University of Florida, from 2012 to 2013. He was an Assistant and an Associate Research Professor of Fire and Unmanned Systems at the Desert Research Institute (DRI), Reno, NV, USA, from 2013 to 2021, and the Deputy Director of the Division of Atmospheric Sciences from 2019 to 2021 at DRI. He is currently a Supervisory Research Biologist at the U.S. Forest Service Pacific Wildland Fire Sciences Laboratory, Seattle, WA, USA. His research interests include systems and disturbance ecology, technology development and applications, and fire science.

Dr. Watts is a member of the Association for Fire Ecology and the International Association for Wildland Fire.



Mohamad Alipour (Member, IEEE) received the B.Sc. and M.Sc. degrees in civil engineering from Tehran Polytechnic, Tehran, Iran, in 2008 and 2011, respectively, and the Ph.D. degree in civil engineering from the University of Virginia, Charlottesville, VA, USA, in 2019.

He was a Post-Doctoral Associate with the Department of Civil and Environmental Engineering at UCLA, Los Angeles, CA, USA, from 2020 to 2022. He is currently a Research Assistant Professor with the Department of Civil and Environmental Engineering, University of Illinois Urbana–Champaign (UIUC), Urbana, IL, USA, and a Faculty Fellow of the National Center for Supercomputing Applications (NCSA), Urbana. His research interests include remote sensing and nondestructive evaluation, digital twins, and robotic infrastructure condition assessment.

Dr. Alipour is currently a member of the American Society of Civil Engineers (ASCE) and an Associate Editor for the *Journal of Nondestructive Evaluation*.